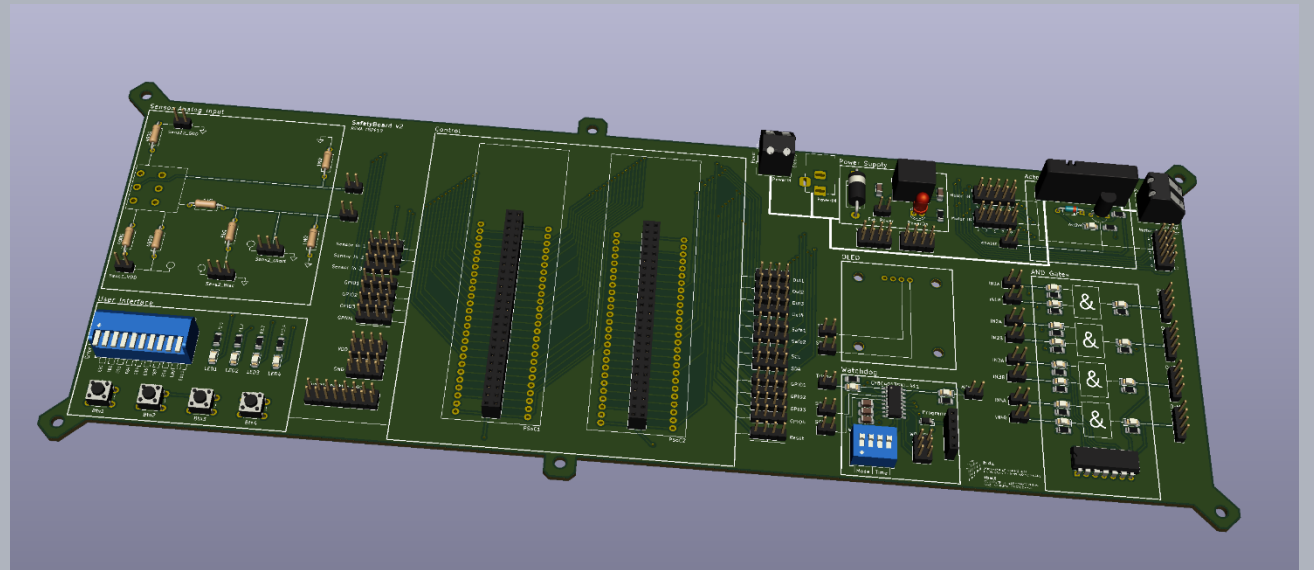


Safety in Embedded Control Systems

System Level Architecture

Prof. Dr.-Ing. Peter Fromm



Content

- Starting point: Safety Requirements
- Technical Safety Concept
- Component Reliability
- Architectural Safety Concepts
- Error detection and Handling
- Analytical Assessment Methods

Last Lecture: Tolerable Risk

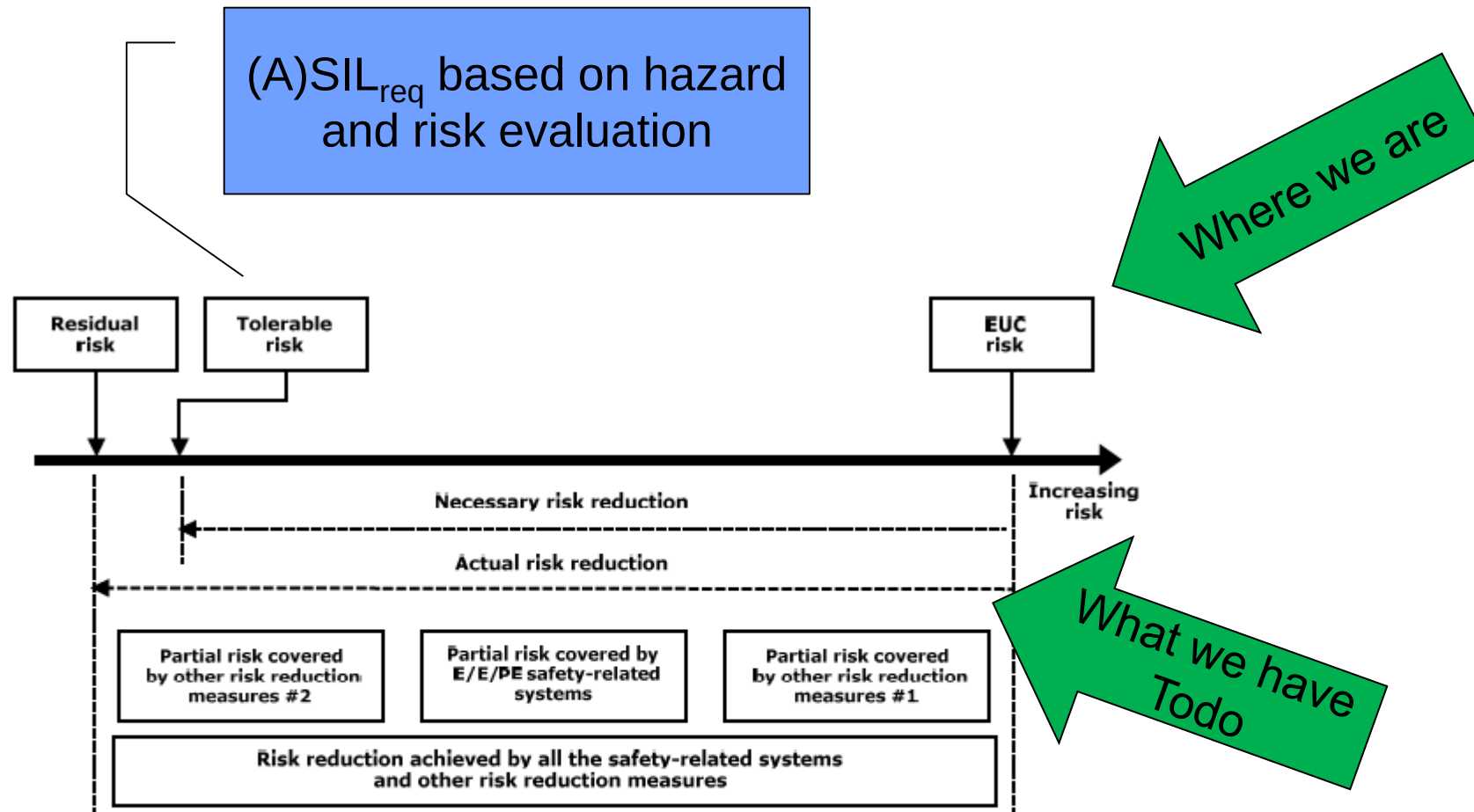
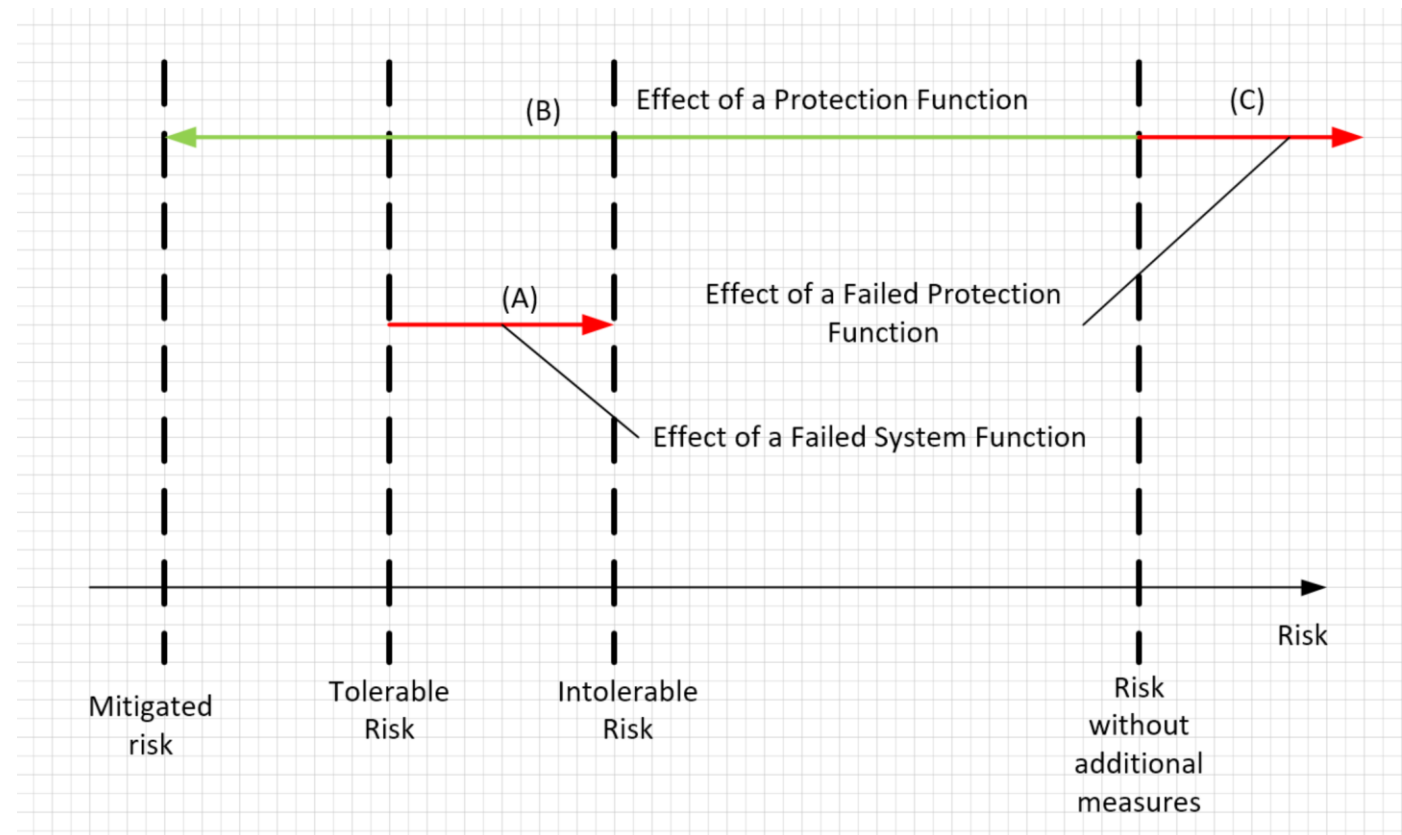


Figure A.1 – Risk reduction – general concepts (low demand mode of operation)

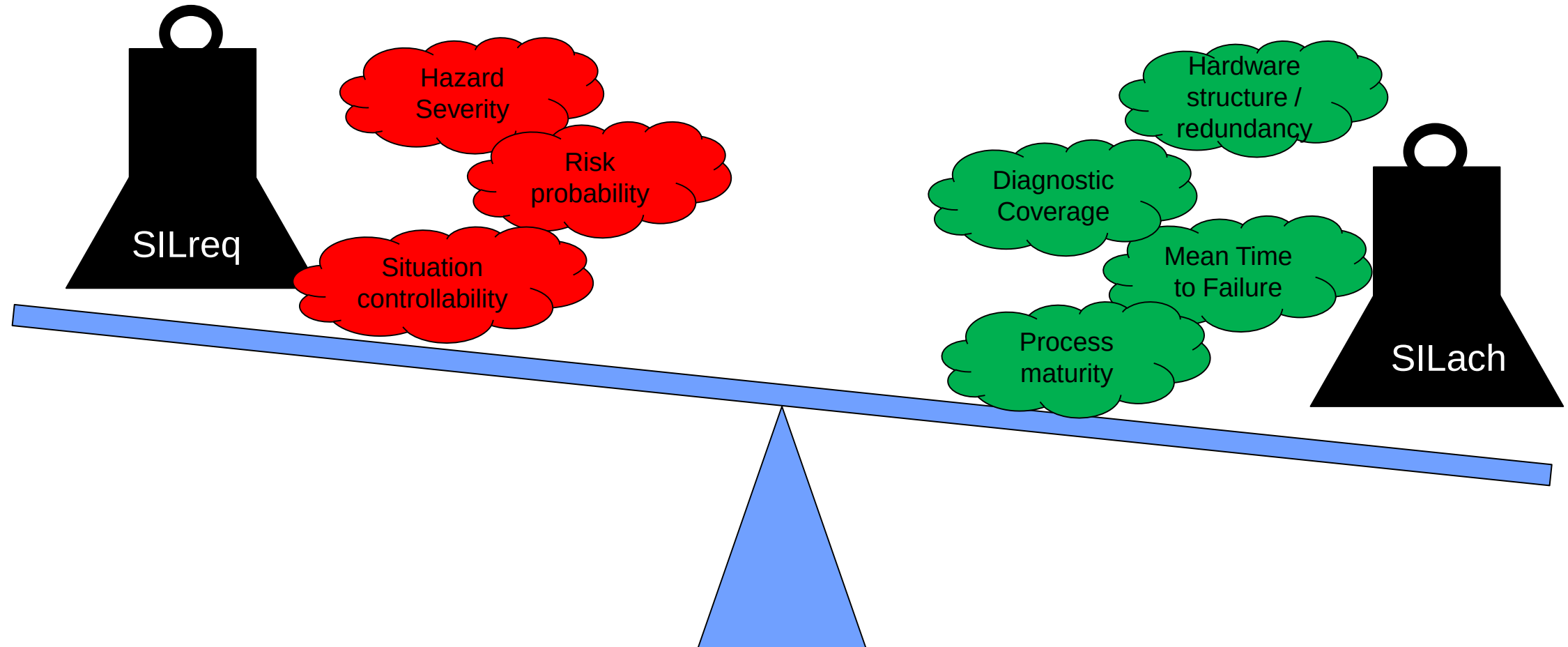
A slightly extended functional view...

- Additional protection functions (e.g. emergency off by a light barrier sensor in front of a dangerous machine) reduce the risk of the overall system
- Safe system functions are required to operate the system in a safe way (e.g. MMI setting the machine speed)
- Both have to be implemented in a safe way in order to avoid harm!



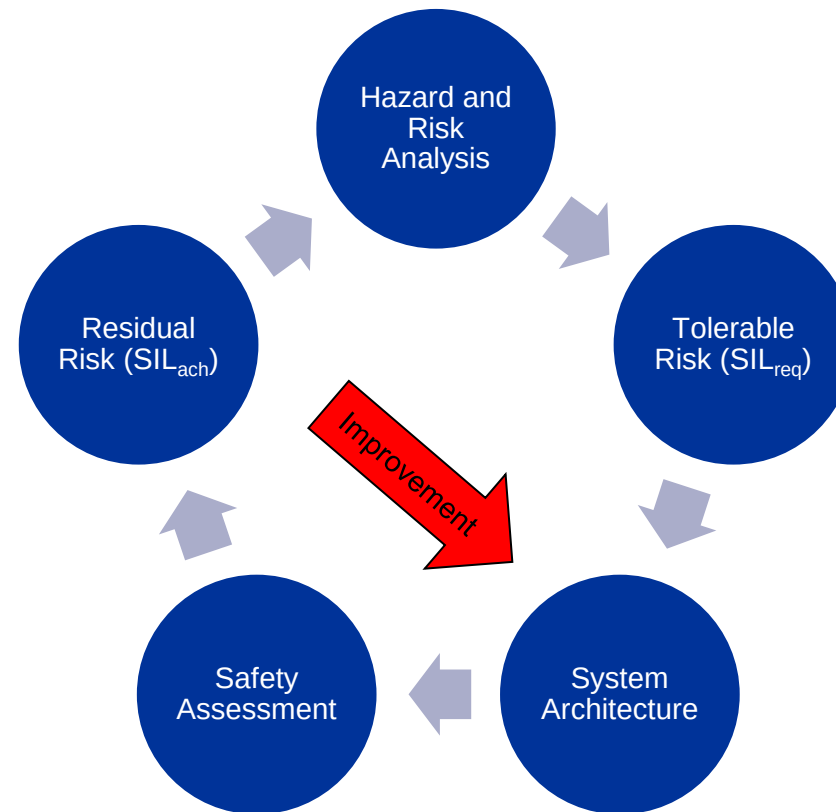
SILrequired versus SILachieved

Although details may differ per standard, the big picture is the following:

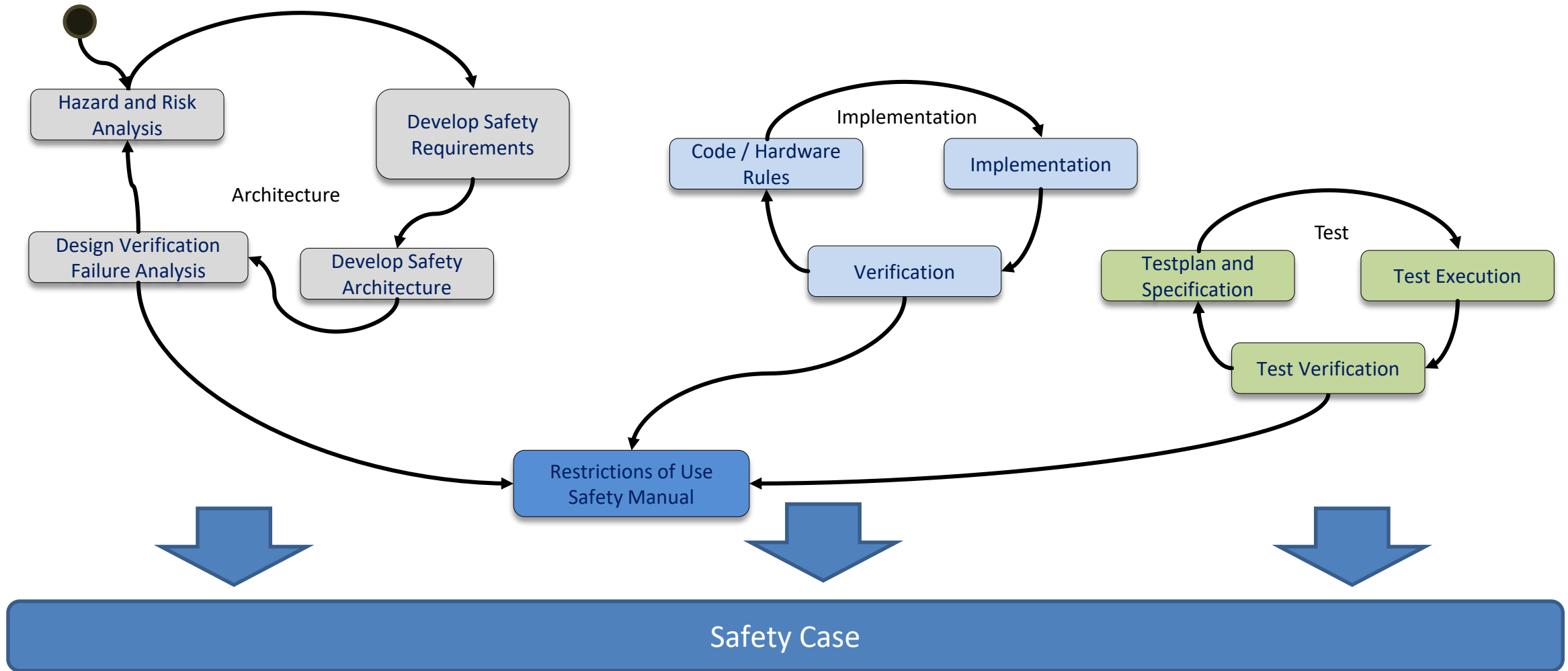


EUC risk and Residual Risk

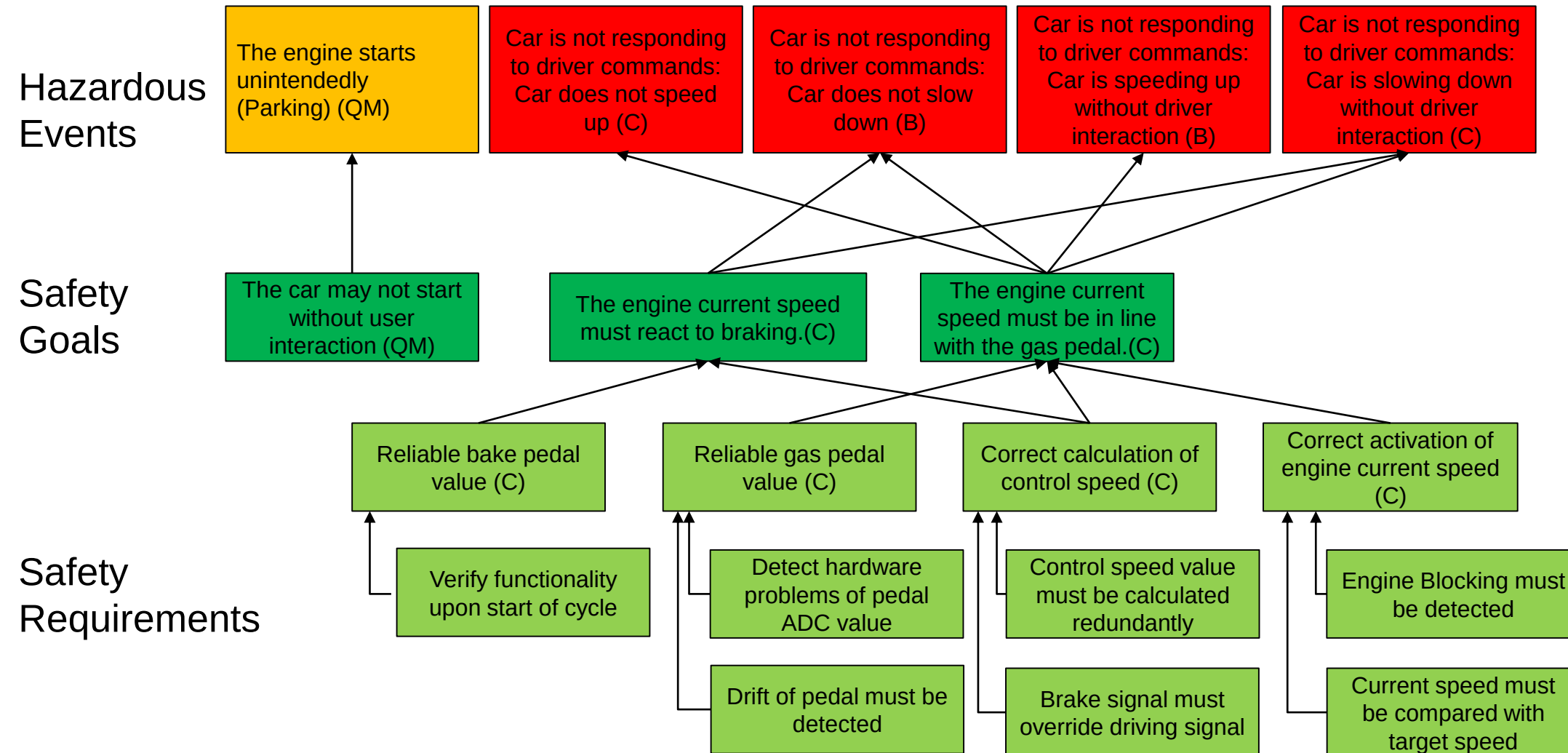
- As we have not developed our system yet, we do not know the EUC risk.
- On the other hand, if we develop our system in such a way, that the Residual Risk is tolerable, we have achieved our goal.



The complete cycle



Overview of a subset of Safety Goals and Safety Requirements (first draft)

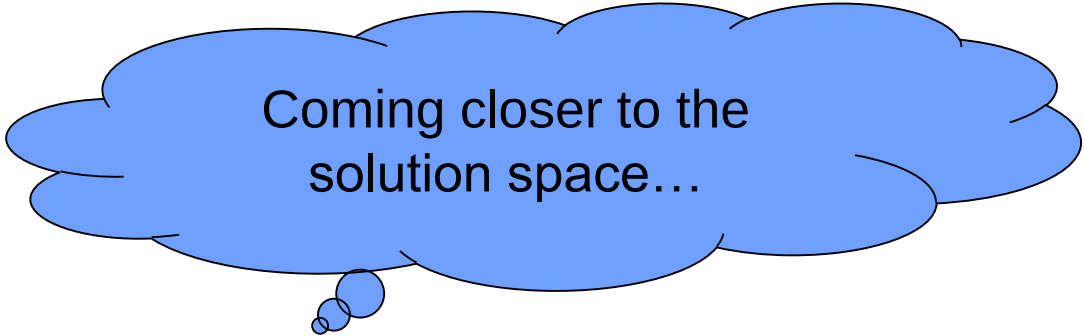


Result of the Hazard and Risk Analysis: A set of Safety Requirements and the required ASIL System Level

Id	Requirements	ASIL _{req}
SR1	The brake signal must be reliable	C
SR2	The gas pedal signal must be reliable	C
SR3	The car may only start after an explicit ignition sequence	QM
SR4	The control speed value must be correctly calculated	C
SR5	The engine currentspeed must be in line with the controlspeed value	C
SR6	The system status must be shown to the driver	A
SR7	The system integrity must be supervised	C

Result of the Hazard and Risk Analysis: A set of Safety Requirements and the required ASIL Broken down into more detailed software and hardware requirements

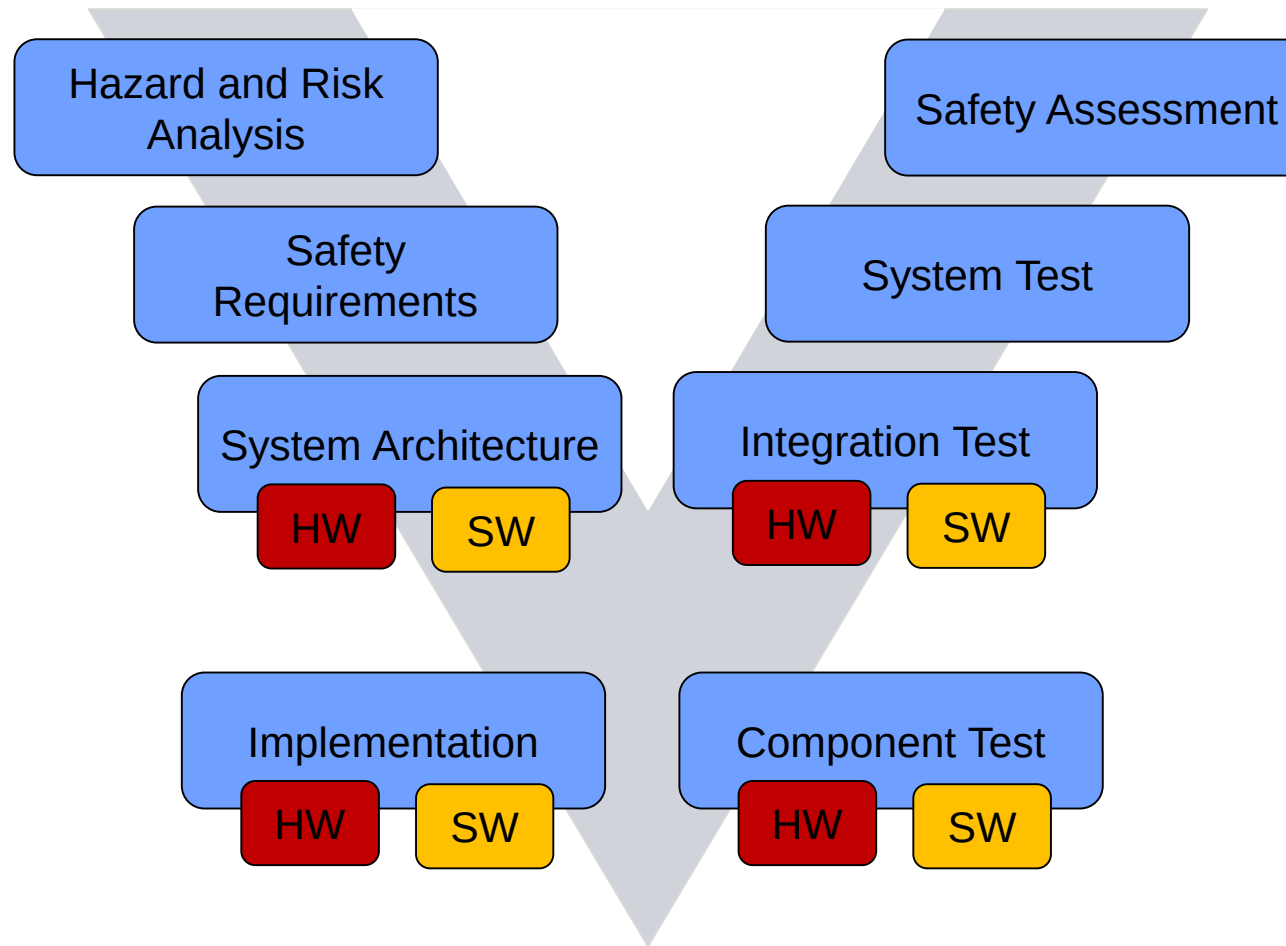
System Level: The brake signal must be reliable



Coming closer to the solution space...

Id	Requirements	ASIL _{req}
SR1.1	HW: Provide 1002 brake signal	C
SR1.2	HW: Provide diagnostics to detect broken connections / short circuits	C
SR1.3	SW: Check correct brake pedal functionality upon power up	C
SR1.6+4	SW: Ensure reliable reading of pedal state	C

Safety requires system level perspective



Hardware / Software Interface

Hardware faults may result in software malfunctions

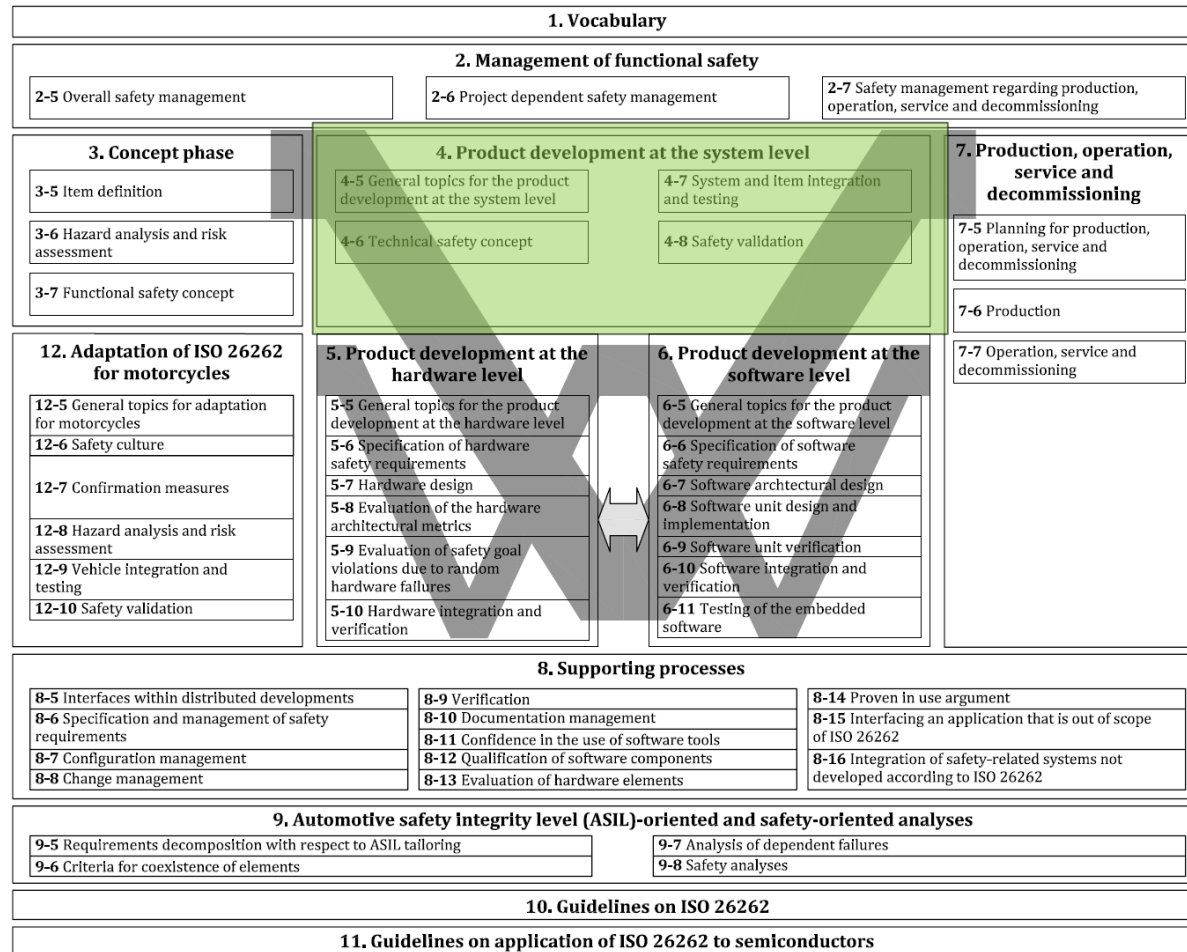
Software can detect hardware faults

Software fault: may not be operational anymore

External hardware must still be able to transition the system into a safe state

System perspective required!

ISO26262 – relevant chapter: 4. Product development at the system level



Focus

- Technical Safety Concept
- System and Item Integration and Testing
- Safety Validation

Figure 1 — Overview of the ISO 26262 series of standards

Technical Safety Concept

6.1 Objectives

The objectives of this clause are:

- a) to specify technical safety requirements regarding the functionality, dependencies, constraints and properties of the system elements and interfaces needed for their implementation;
- b) to specify technical safety requirements regarding the safety mechanisms to be implemented in the system elements and interfaces;
- c) to specify requirements regarding the functional safety of the system and its elements during production, operation, service and decommissioning;
- d) to verify that the technical safety requirements are suitable to achieve functional safety at the system level and are consistent with the functional safety requirements;
- e) to develop a system architectural design and a technical safety concept that satisfy the safety requirements and that are not in conflict with the non-safety-related requirements;
- f) to analyse the system architectural design in order to prevent faults and to derive the necessary safety-related special characteristics for production and service; and
- g) to verify that the system architectural design and the technical safety concept are suitable to satisfy the safety requirements according to their respective ASIL.

6.2 General

The **technical safety concept** is an aggregation of the technical safety requirements and the corresponding system architectural design that provides rationale as to why the system architectural design is suitable to fulfil safety requirements resulting from activities described in ISO 26262-3 (with consideration of non-safety requirements) and design constraints.

The **technical safety requirements** specify the technical implementation of the functional safety requirements at their respective hierarchical level; considering both the item definition and the system architectural design, and addressing the detection of latent failures, fault avoidance, safety integrity and operation and service aspects.

The **system architectural design** is the selected system-level solution that is implemented by a technical system. The system architectural design aims to fulfil both, the allocated technical safety requirements and the non-safety requirements.

System development can be performed iteratively.

There is no such thing as THE ONE AND ONLY SAFETY ARCHITECTURE.
Instead we have to argue why our design is safe.

Technical Safety Concept

Specification of the technical safety requirements

- We have identified safety goals and requirements, which now must be translated into technical solutions.
- ISO26262: *“The technical safety requirements shall specify the stimulus response of the system that affects the achievement of safety requirements. This includes the combinations of relevant stimuli and failures with each relevant operating mode and defined system state.”*
- Example: The gas pedal signal must be reliable
 - The gas pedal position will be measured using 2 different sensors.
 - Furthermore diagnostic resistors will be added to detect short circuits or broken connections.
 - If the validity of at least one channel is ensured, a warning light will turn on and the max speed will be limited to 50%
 - If the validity of both channels is uncertain (e.g. if the values differ for more than xyz % OR are not updated once every xyz ms) a warning light will blink and the engine will stop (no active braking).

Technical Safety Concept

Safety mechanisms

6.4.2 Safety mechanisms

6.4.2.1 The technical safety requirements shall specify the safety mechanisms that detect faults and prevent or mitigate failures present at the output of the system that violate the functional safety requirements (see ISO 26262-3:2018, Clause 7) including:

- a) the safety mechanisms related to the detection, indication and control of faults in the system itself;

NOTE 1 This includes the system self-monitoring to detect random hardware faults and, if appropriate, to detect systematic faults.

NOTE 2 This includes safety mechanisms for the detection and control of communication channel failures (e.g. data interfaces, communication buses, wireless radio link).

NOTE 3 Safety mechanisms can be specified with respect to the appropriate level within the system architecture.

- b) the safety mechanisms related to the detection, indication and control of faults in other external elements that interact with the system;

EXAMPLE External devices include other electronic control units, power supplies or communication devices.

- c) the safety mechanisms that contribute to the system achieving or maintaining the safe state of the item;

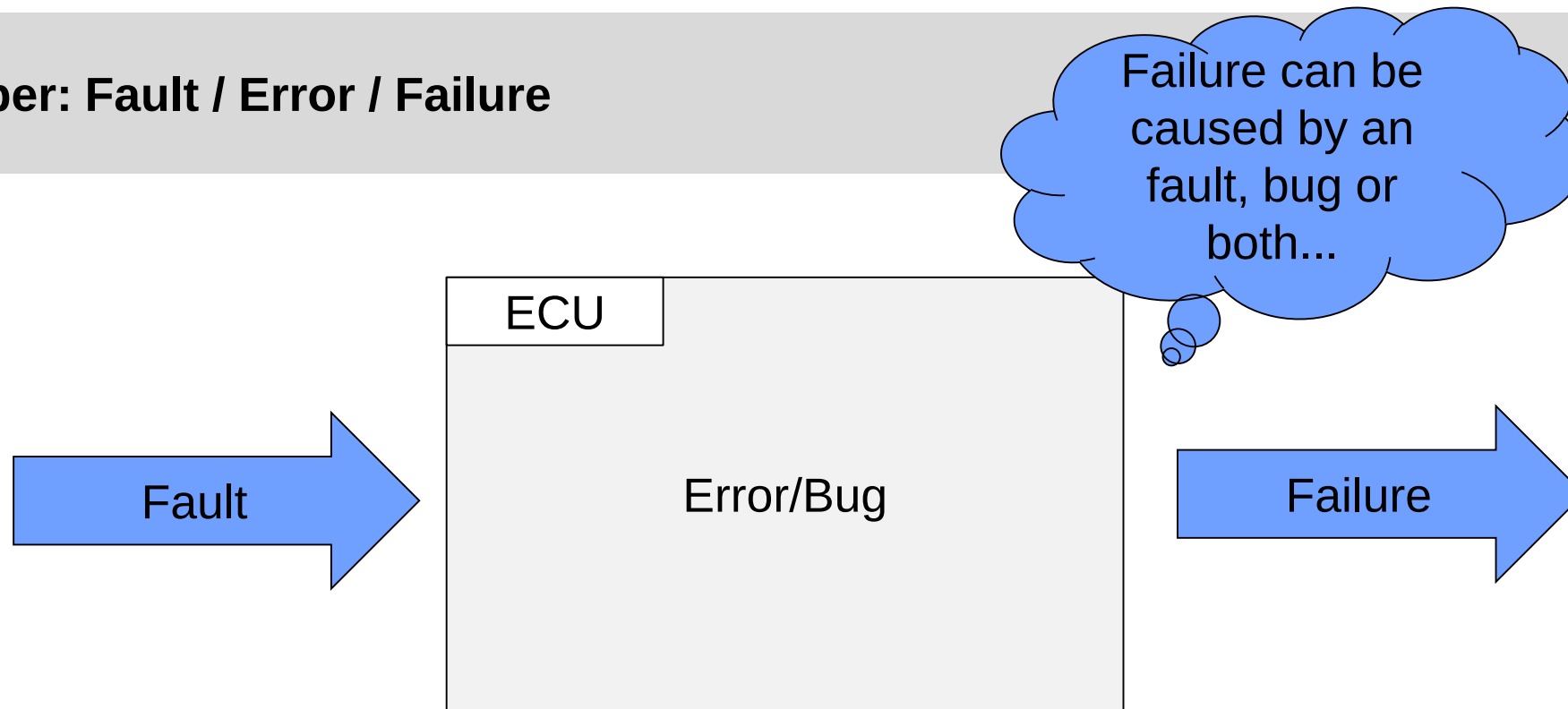
NOTE 4 This includes arbitration in the case of multiple control requests from safety mechanisms.

- d) the safety mechanisms to define and implement the warning and degradation strategy; and

- e) the safety mechanisms that prevent faults from being latent.

NOTE 5 These safety mechanisms are usually related to self-tests that take place during power up (pre-drive checks), as in the case of measures a) to d), during operation, during power-down (post-drive checks), and as part of maintenance.

Remember: Fault / Error / Failure



e.g.

- Wrong sensor Value
- Wrong user Input
- Transmission error
- ...

e.g.

- Logical error
- Data overflow
- Race condition
- ...

e.g.

- Wrong data
- Wrong actor motion
- Wrong protocol
- Wrong HMI output...

Main Requirements for a Safety Architecture



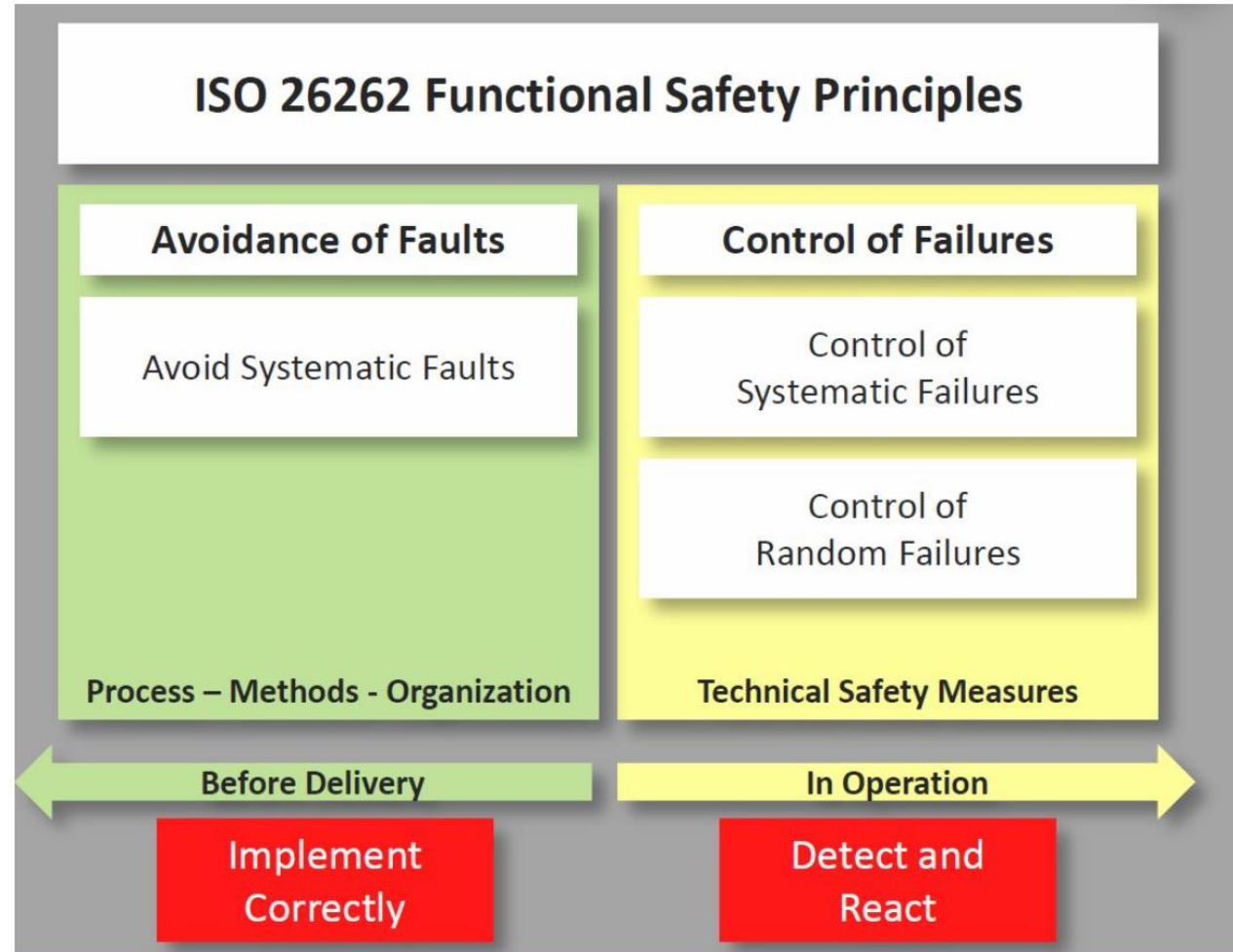
Avoid Dangerous Failures

Reliable detection and handling of stochastic faults and errors (focus single faults)
E.g. hardware failures, communication errors,...

Reduction / avoidance of systematic errors.
E.g. programming bugs

In case of a critical failure: Transition into a safe state.

Another perspective



University of Toronto, CSC2125, Lecture 3: Safety Analysis

Typical Safety Mechanisms

Component Reliability

MTTF

Process

Fault Detection

Diagnostic Coverage

Error Handling

Architecture

Redundancy

Freedom from Interference

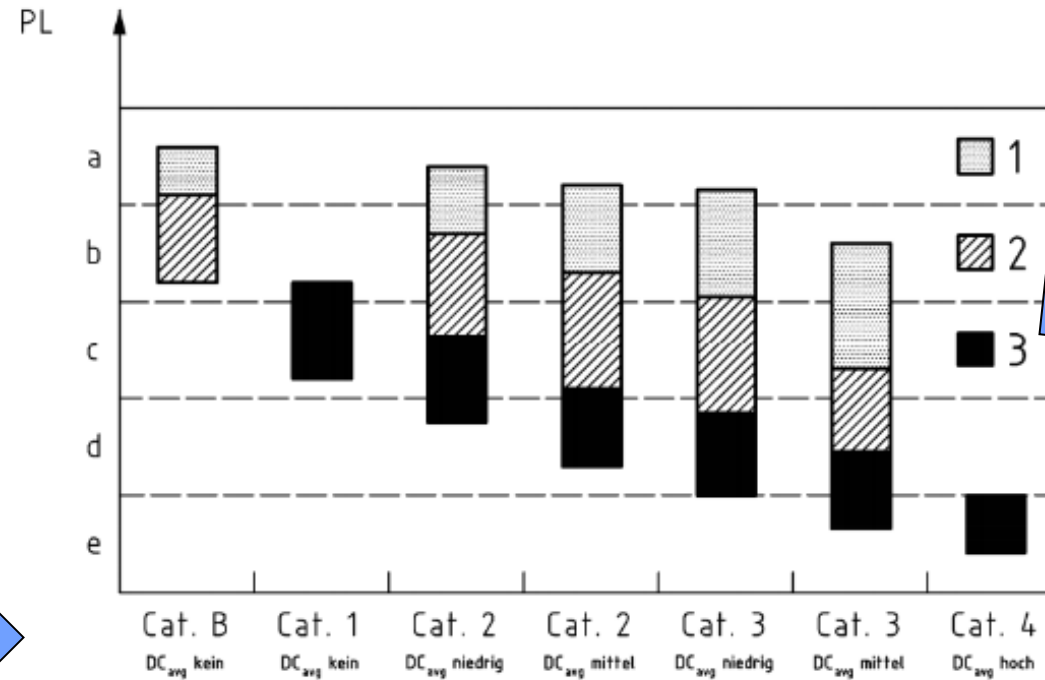
Common Cause Failure

Safe State

Machine Industry: ISO13849

Resulting SIL
(Performance Level)

Architecture



Legende

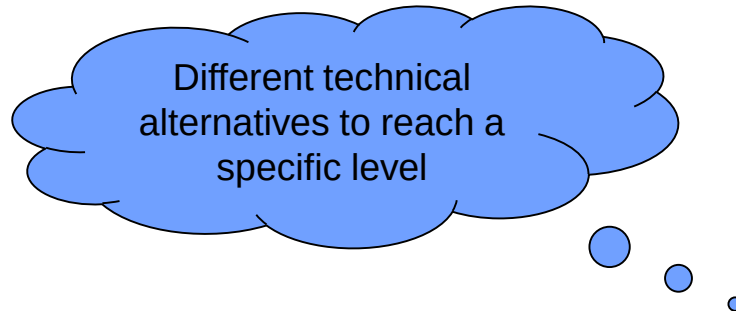
PL Performance Level

- 1 $MTTF_d$ jedes Kanals = niedrig
- 2 $MTTF_d$ jedes Kanals = mittel
- 3 $MTTF_d$ jedes Kanals = hoch

Component Reliability

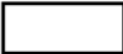
Diagnostics

Tractors: ISO 25119



AgPL		SRL			
a	1	B	B	B	B
b	2	1	B	B	B
c		2	1	1	1
d			2	2	2
e					3
	Cat B	Cat 1	Cat 2	Cat 3	Cat 4
	DC low	DC med	DC med	DC med	DC high

Key

	low $MTTF_{DC}$
	medium $MTTF_{DC}$
	high $MTTF_{DC}$

Note:

AgPL == SIL

Cat B, 1, 2, 3, 4 describe the hardware architecture (redundancy)

SRL stands for software reliability level and describes the requirements for the software process.

Component Reliability - Hardware

- Focus: stochastic errors
- Failure rate λ [FIT]
- Failure in Time = $1 \times 10^{-9} \text{h}$ / 1 failure every 114.000 years approx.
- Based on experience / measurement data
- Factors like temperature, overvoltage etc. must be considered
- Standard components: SN-29500
- Company specific FMEDA documents
- Microcontroller: Use Case specific evaluation depending on applied hardware safety features

Tabelle 5 Ausfallraten für anwendungsspezifische Schaltkreise (ASICs)
Table 5 Failure rates for application-specific integrated circuits (ASICs)

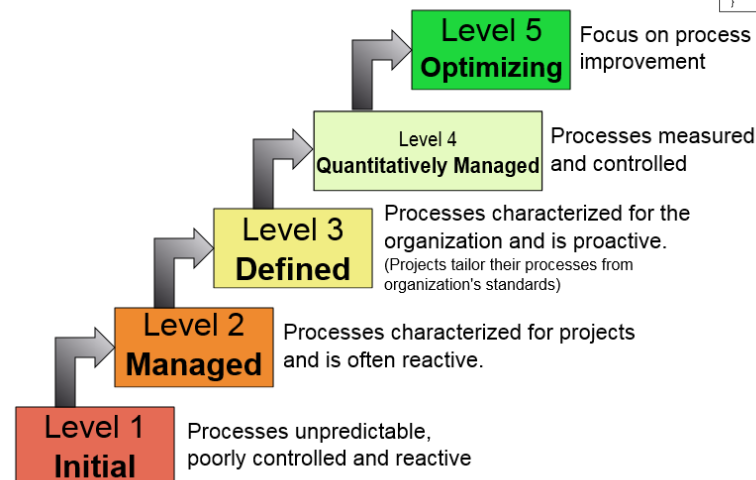
Gatteranzahl / No. of gates Transistoranzahl / No. of transistors		Integrationsgrad / Degree of integration						λ_{ref} in FIT in °C
		MSI/LSI 10 – 1k	VLSI >1k - 10k	ULSI >10k - 100k	SLSI >100k - 1M	>1M - 10M	>10M - 100M	
		50 – 5k	>5k - 50k	>50k - 500k	>500k - 5M	>5M - 50M	>50M - 500M	
λ_{ref} in FIT								
ASICs, Full Custom, Gate Arrays, Telecom ICs, A/D-Converters ¹⁾								
Bipolar	TTL	20	30	-	-	-	-	55
	ECL	40	60	-	-	-	-	70
	HV (>50 Volt) ^{2) 3)}	-	40	-	-	-	-	80
NMOS		25	30	-	-	-	-	55
CMOS	digital,	20	25	-	-	-	-	55
BICMOS	analog / mixed ³⁾	-	-	60	70	80	-	70
		-	-	-	-	-	120	80
	HV (>50 Volt) ^{2) 3)}	-	30	-	-	-	-	75
Programmable ASICs (PLD) ⁵⁾ nicht löschar / non-erasable								
Bipolar	TTL	50	-	-	-	-	-	80
	ECL	70	-	-	-	-	-	85
CMOS	(antifuses)	-	80	90	120	-	-	80
Programmable ASICs (PLD) ⁵⁾ löschar / erasable								
MOS	RAM basis	-	60	80	100	(150)	(200)	80
CMOS	EPROM basis ⁶⁾	40	-	-	-	-	-	70
		-	50	80	-	-	-	80
	EEPROM basis	40	-	-	-	-	-	70
	FLASH-EPROM	-	60	90	120	(160)	-	80
1 FIT=1x10⁻⁹ 1/h; (Ein Ausfall pro 10⁹ Bauelementestunden) Für Bauelemente ohne ausreichende Einsatzserfahrungen sind die Ausfallratenwerte eingeklammert. Bei der Verwendung von nackten Chips sind die Ausfallraten mit einem Faktor von bis zu 2 zu multiplizieren, wenn keine eigenen Erfahrungen in der Aufbautechnik vorliegen. 1) Monolithische Technologie (keine Module) 2) Die Ausfallraten gelten für den typischen Einsatzfall: Vollast während 10% der Betriebszeit. 3) Für driftpfindliche Schaltungen ist der Driftsensitivitätsfaktor λ_D zu berücksichtigen (siehe Abschnitt 4.3). 4) Ausfallrate in Abhängigkeit von Komplexität festlegen. 5) Bei PLDs werden nur die Gatter der logischen Funktion gerechnet. 6) Auch gültig für nicht löscharbare Versionen im Plastikgehäuse. 1 FIT equals one failure in 10⁹ component hours Failure rates of components for which little operating experience has been gained are given in brackets. For bare chips the indicated failure rates shall be multiplied by a factor of up to two, if no first-hand experience in the design layout techniques is available. 1) Monolithic technology (no modules) 2) The failure rates stated apply to the typical application: full load during 10% of the operating time. 3) In drift-sensitive circuits the drift sensitivity factor λ_D must be taken into account (see Clause 4.3). 4) The failure rate is to be determined depending on the complexity. 5) For PLDs only the gates of the logical functions are considered. 6) Also valid for non-erasable types in plastic packages.								

Source: SN-29500 part 2

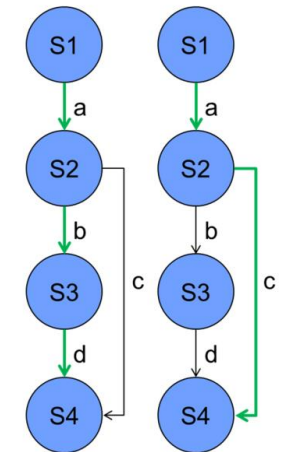
Component Reliability - Software

- Focus: systematic errors
- Process maturity
- Use of models
- Application of design and coding patterns
- Static code analysis and reviews
- Tests and test coverage

Characteristics of the Maturity levels



```
RT_safety_t REMOTE_run_checkJoystickAge()
{
    CARSTATE_data_t carState = CARSTATE_get(&so_carstate);
    if (carState.state_normal == IS_RUNNING) {
        if (JOYSTICK_get_age(&so_joystick) >=
SAFETY_JOYSTICK_TIMEOUT_ERROR) {
            //Stop the car
            return SAFEERROR2;
        }
        //MISRA: A function should have a single point of exit
    }
    else if (JOYSTICK_get_age(&so_joystick) >=
SAFETY_JOYSTICK_TIMEOUT_WARNING) {
        //Slow the car down
        return SAFEERROR1;
    }
    //MISRA: A function should have a single point of exit
    //MISRA: Else if with no following else
    if (carState.state_normal != IS_RUNNING && carState.state_normal
!= IS_PRERUN) {
        JOYSTICK_set_status_invalid(&so_joystick);
    }
    //MISRA: Discarded return value
    return SAFEOK;
}
```

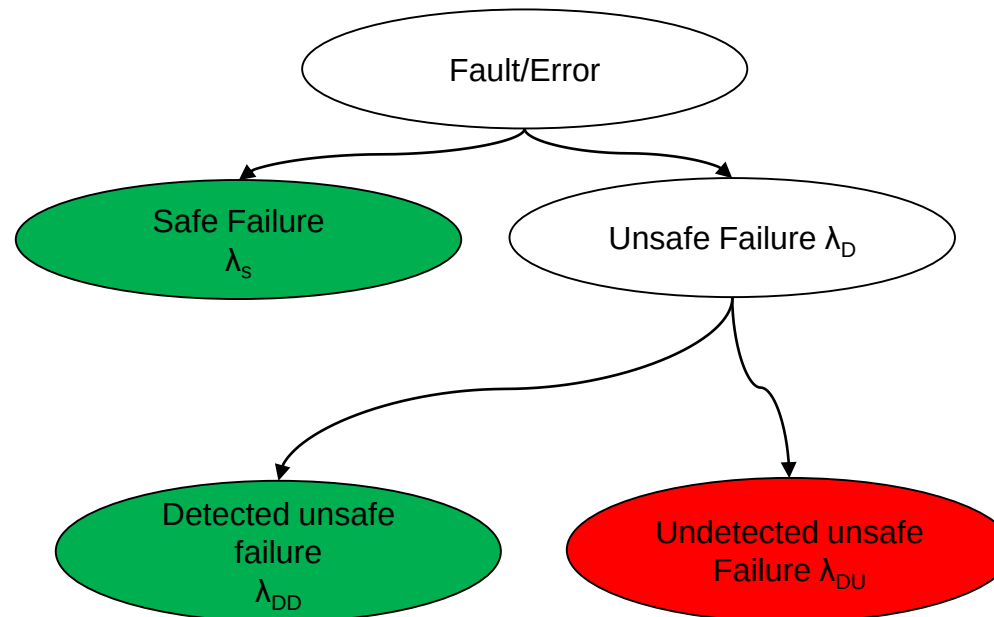


Diagnostic Coverage and Safe Failure Fraction

- The Safe Failure Fraction defines the ration of detected and safely handled failures in relation to the total failures.
- The Diagnostic Coverage is the ratio of detected dangerous failures in relation to the total dangerous failures
- Typical ranges (depending on standard)
 - 99%...100% (High)
 - 90%..99% (Medium)
 - Below 90% (Low)

$$SFF = \frac{\sum \lambda_s + \sum \lambda_{DD}}{\sum \lambda_s + \sum \lambda_{DD} + \sum \lambda_{DU}}$$

$$DC = \frac{\lambda_{DD}}{\lambda_D}$$

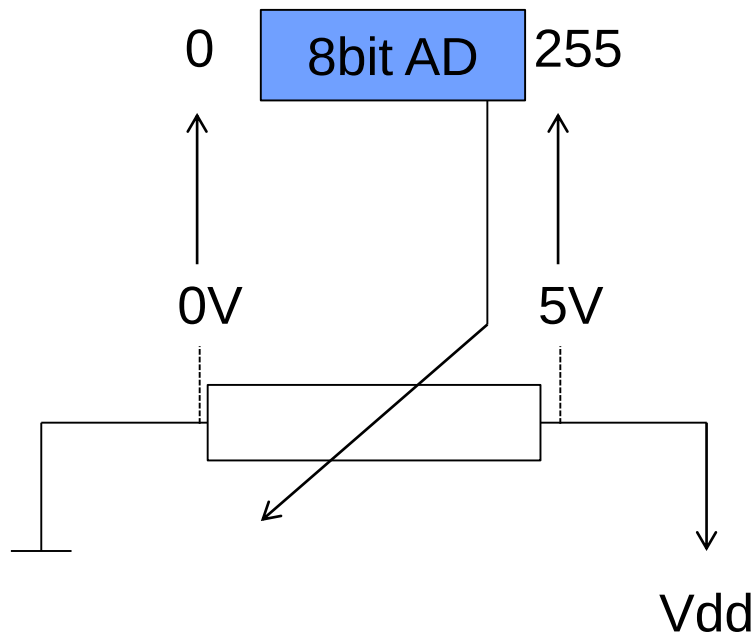


DC values for different standards

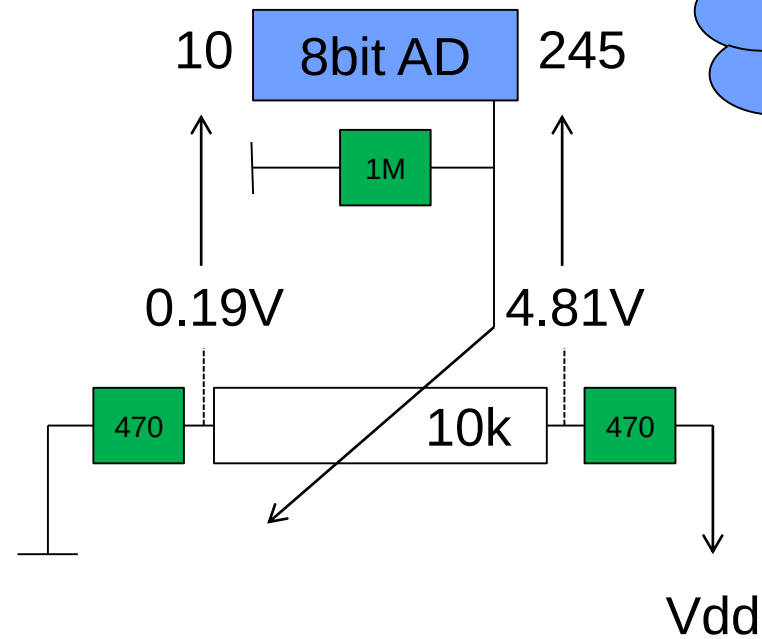
DC	IEC 61508:2	ISO 25119:2	ISO 26262:5
Low	>60%	0..60%	>60%
Medium	>90%	>60%	>90%
High	>99%	>90%	>99%

Diagnostic Coverage – Example Potentiometer

Range: 0..255
Valid Range: 0..255



Range: 0..255
Valid Range: 11..244
Error Range: 0..10 and 245..255



How about
communication?
Software?

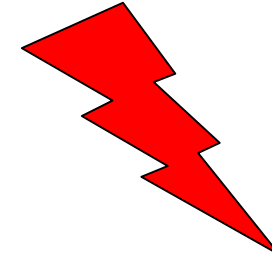
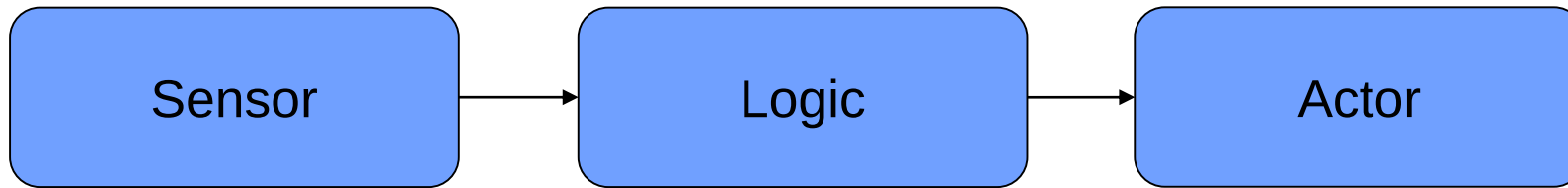


Attention: Short circuits and broken connections can be detected, but not bad signals from the potentiometer itself.

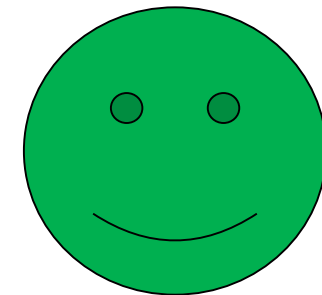
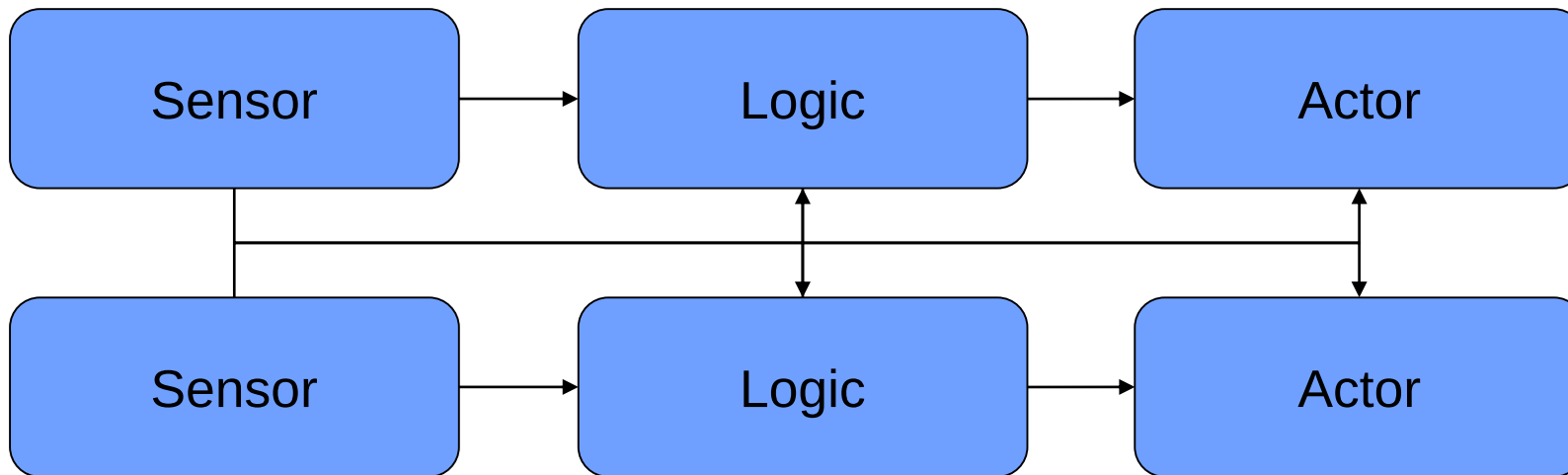
DC Calculation for Input Channel (informal FMEDA)

Id	Error	Dangerous	Probability	Detected
1	Left resistor connection broken			
2	Left resistor short circuit to GND			
3	Left resistor short circuit to VDD			
4	Left resistor drift (<10%)			
5	Left resistor drift (>10%)			
6	MCU Connection broken between resistor and potentiometer			
7	MCU Connection broken between resistor and MCU			
8	Wrong value (<10%) of potentiometer			
9	Wrong value (>10%) of potentiometer			
	...			

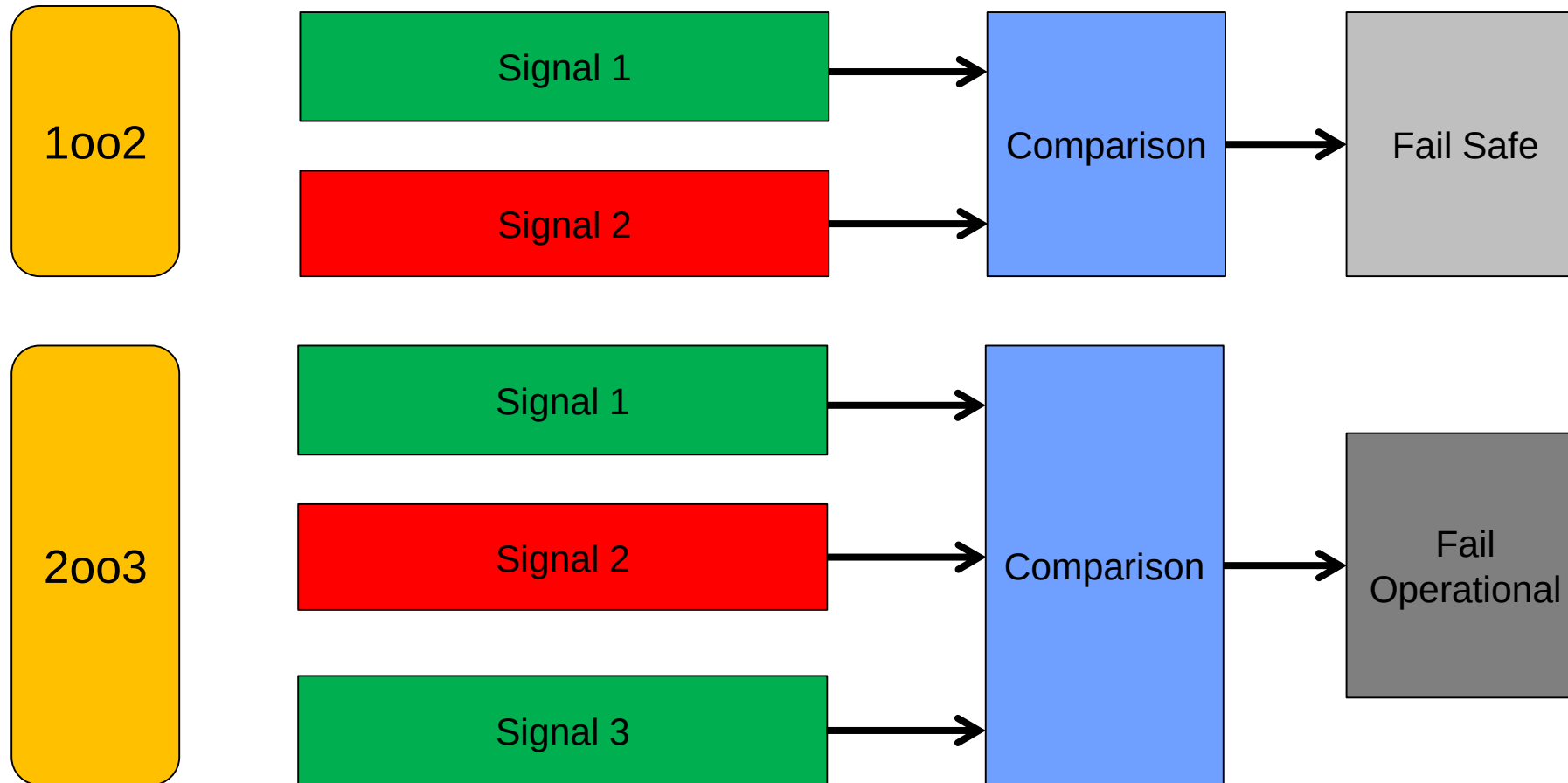
Redundancy / Hardware Fault Tolerance



- A fault / failure in any of the components will lead to a channel failure
- Key idea: data is read and processed over more than one channel



1oo2 and 2oo3

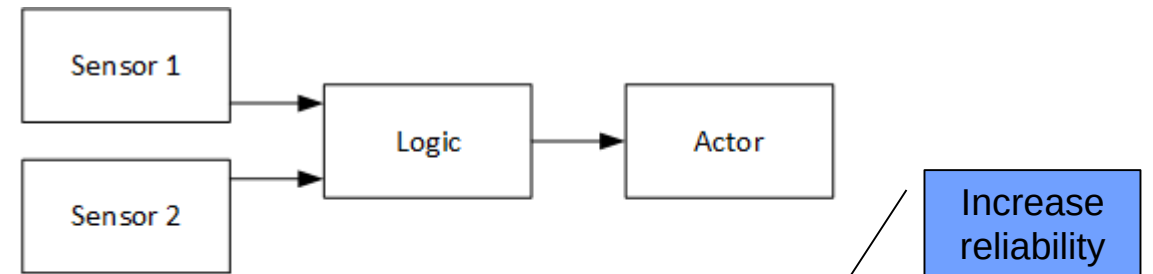
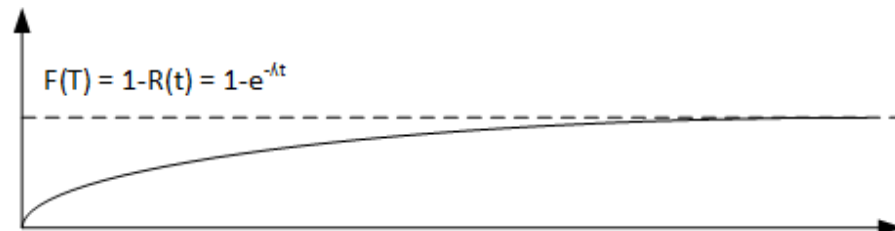
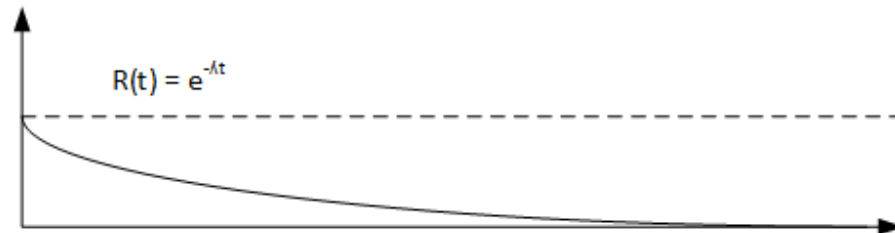


Failure Rate and Hardware Fault Tolerance

Failure Rate: $\lambda(t)$ in FIT

Reliability / Survival Probability: $R(t) = e^{-\lambda t}$

Failure Probability: $F(T) = 1 - R(t) = 1 - e^{-\lambda t}$



$$F_{S1+S2}(t) = F_{S1}(t) \times F_{S2}(t)$$

$$R_{S1+S2}(t) = R_{S1}(t) + R_{S2}(t) - R_{S1}R_{S2}(t)$$

Parallel components: Failure Rates are multiplied

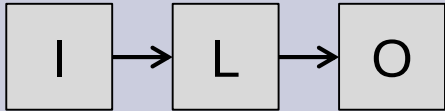
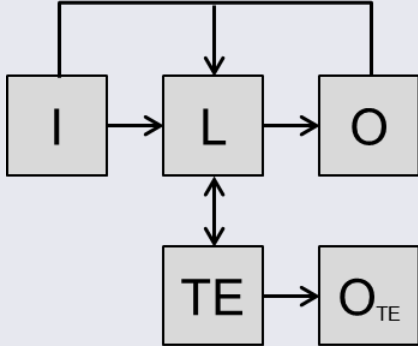
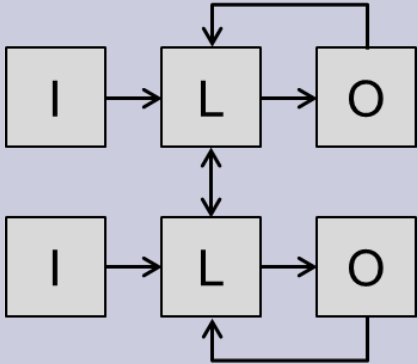
$$F_{ch}(t) = F_{S1+S2}(t) + F_L(t) + F_A(t)$$

$$R_{ch}(t) = R_{S1+S2}(t) \times R_L(t) \times R_A(t)$$

Serial components: Failure Rates are added

Decrease reliability

Architecture Evaluation according ISO13849

Category	Structure	Description
B / 1		• Single Channel • No Test Function • Proven in Use Hardware
2		• Single Channel • External Test Function • Safe State reachable through Test Function • Individual path are secured against common cause failures
3 / 4		• Redundant channel • Crosswise supervision • Individual path are secured against common cause failures

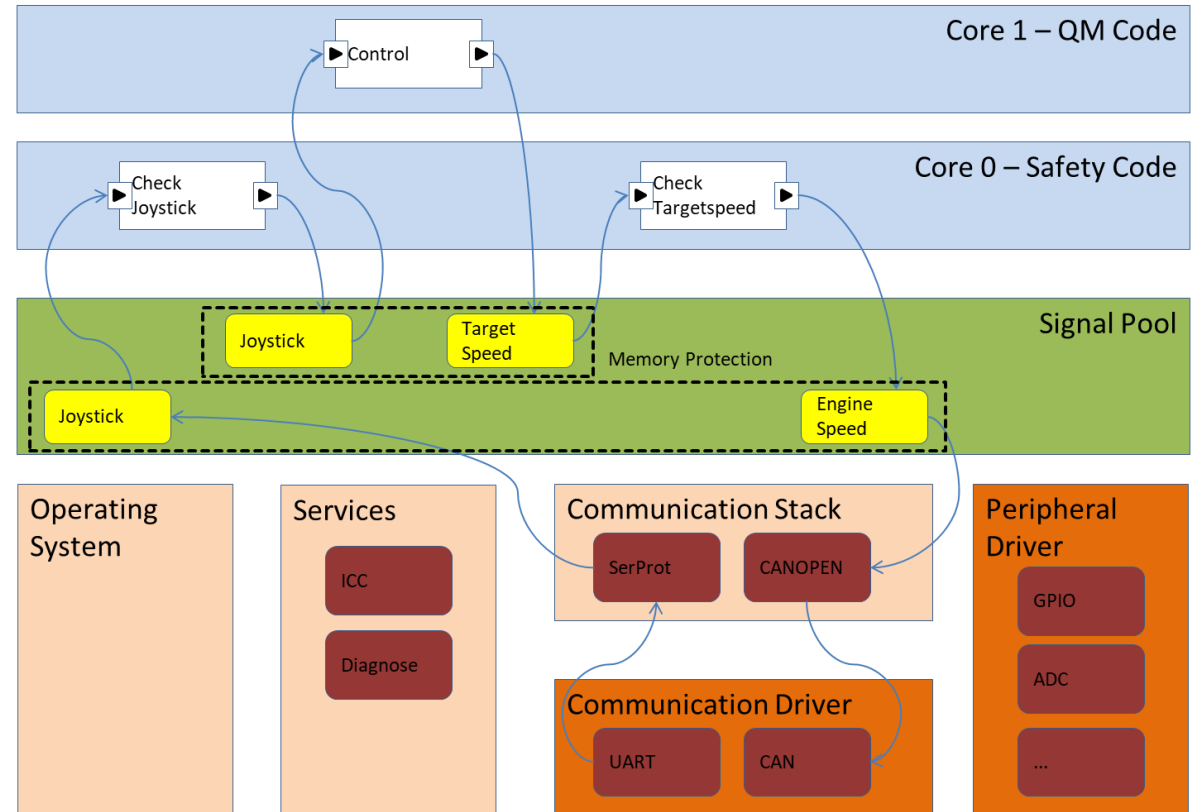
Freedom from Interference

Errors in QM (or SIL) code may not affect the correct execution of other safety functions. I.e. faults should not lead to a uncontrollable degrading of the system integrity.

- Memory domain (e.g. wild pointers or global variables)
- CPU Time Domain (One task blocks other tasks)
- Peripherals (One function reconfigures or blocks peripheral pins)

Solutions

- Redundant controllers (expensive)
- MPU
- Alive / Deadline Monitoring



Common Cause Failures

- Closely related with redundancy - a single failure may not have a common negative impact on hardware fault tolerance.
- Physical separation of power domains
- Diversity of components
- Diversity of technology
 - E.g. VADC and Sigma-Delta
 - Different software developers

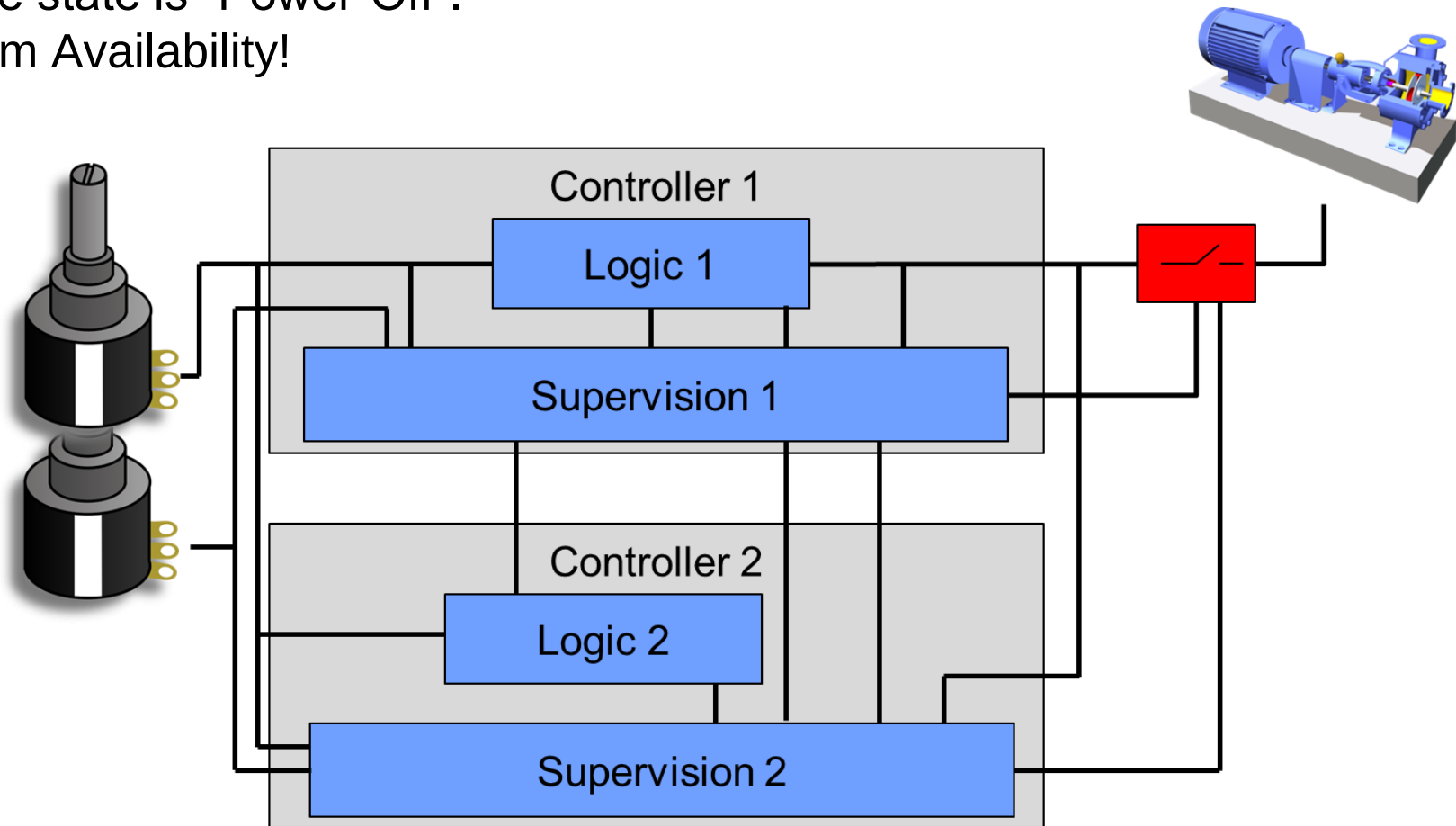
Tabelle F.1 — Verfahren zur Punktevergabe und Quantifizierung für Maßnahmen gegen CCF

Nr.	Maßnahme gegen CCF	Punktzahl
1	Trennung/Abtrennung Physikalische Trennung zwischen den Signalpfaden, z. B.: — Trennung der Verdrahtung/Verrohrung; — Erkennen von Kurzschlüssen und Unterbrechungen in Kabeln durch dynamische Prüfung; — getrennte Abschirmung des Signalpfades jedes Kanals; — ausreichende Luft- und Kriechstrecken auf gedruckten Schaltungen.	15
2	Diversität Unterschiedliche Technologien/Gestaltung oder physikalische Prinzipien werden verwendet, z. B.: — der erste Kanal elektronisch oder programmierbar elektronisch und der zweite Kanal elektromechanisch fest verdrahtet; — unterschiedliche Initiierung der Sicherheitsfunktion für jeden Kanal (z. B. Position, Druck, Temperatur), und/oder digitale und analoge Messung von Variablen (z. B. Abstand, Druck oder Temperatur) und/oder Bauteile von unterschiedlichen Herstellern.	20
3	Gestaltung/Anwendung/Erfahrung	
3.1	Schutz gegen Überspannung, Überdruck, Überstrom, Übertemperatur, usw.	15
3.2	Verwendung bewährter Bauteile.	5
4	Beurteilung/Analyse Für jedes Teil von sicherheitsbezogenen Teilen eines Steuerungssystems wurde eine Ausfallarten- und Effekt-Analyse durchgeführt und deren Ergebnisse berücksichtigt, um Ausfälle infolge gemeinsamer Ursache bei der Gestaltung zu vermeiden.	5
5	Kompetenz/Ausbildung Ausbildung der Konstrukteure, um die Gründe und Auswirkungen von Ausfällen infolge gemeinsamer Ursache zu verstehen.	5
6	Umgebung Für elektrische/elektronische Systeme, Verhindern von Verunreinigungen und elektromagnetischen Störungen (EMV) zum Schutz vor Ausfällen infolge gemeinsamer Ursache entsprechend den einschlägigen Normen (z. B. IEC 61326-3-1) 6.1 Fluidische Systeme: Filtrierung des Druckmediums, Verhinderung von Schmutzeintrag, Entwässerung von Druckluft, z. B. in Übereinstimmung mit den Anforderungen des Herstellers für die Reinheit des Druckmediums ANMERKUNG Bei kombinierten fluidischen und elektrischen Systemen sollten beide Aspekte berücksichtigt werden. 6.2 Andere Einflüsse: Berücksichtigung der Anforderungen hinsichtlich Unempfindlichkeit gegenüber allen relevanten Umgebungsbedingungen wie Temperatur, Schock, Vibration, Feuchte (z. B. wie in den zutreffenden Normen festgelegt).	25
	Gesamt	[max. erreichbar 100]
Gesamtpunktzahl		Maßnahmen, um CCF^a zu vermeiden
65 oder besser		Anforderungen erreicht
Kleiner als 65		Verfahren gescheitert ⇒ Auswahl zusätzlicher Maßnahmen

^a Wenn technische Maßnahmen nicht relevant sind, können die Punkte der rechten Spalte bei der ausführlichen Berechnung berücksichtigt werden.

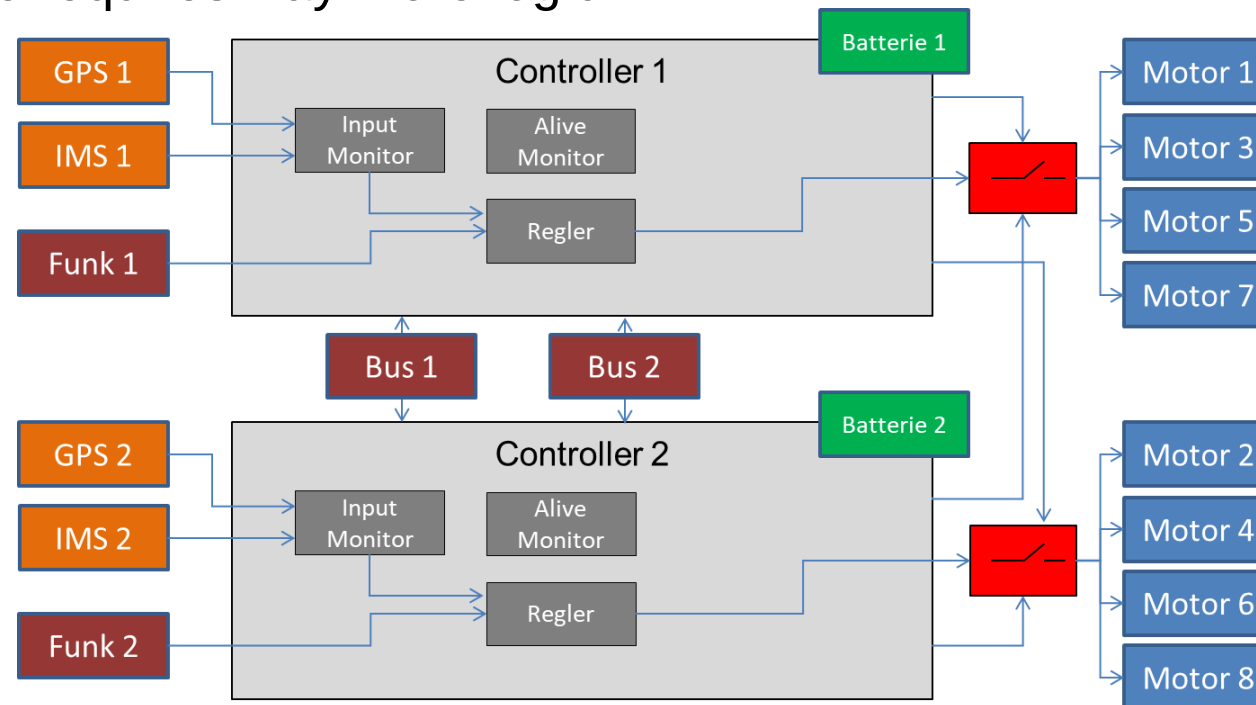
Safe State / Failsafe

- A safe state of a system is an operational state, in which hazardous situations are avoided.
- A safe state must be reachable through a simple and reliable transition / operation.
- Often, such a safe state is “Power-Off”.
- Challenge: System Availability!



Safe State / Fail Operational

- In certain situations, simply turning off the device is no option (e.g. plane flying)
- In this case, an operational state (often with limited functionalities) needs to be reached, which allows a safe operation of the device until a safe “off” state can be reached (e.g. plane on the ground, car in the garage,...)
- The implementation of a fail operational system is significantly more complex, as the transition into such an operational state requires way more logic.



Fail Safe / Fail Operational Trend

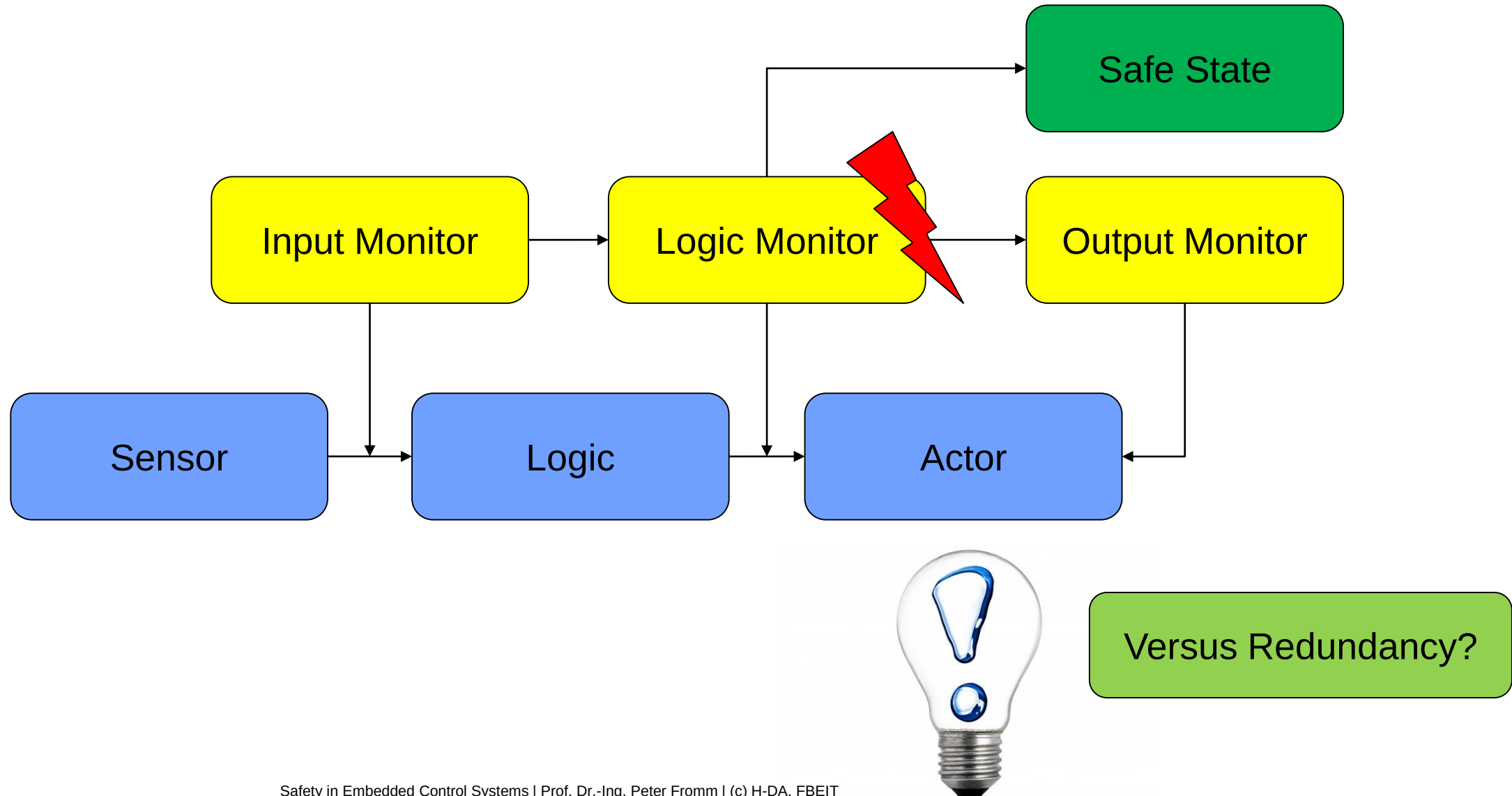
SAE Level	Previous Gen		Current Gen		Next Gen	
	FAIL SAFE		SAFETY & AVAILABILITY		FAIL OPERATIONAL	
	Detect Fault Indicate fault to Safe State System		Detect Fault Indicate fault to Safe State System and recover		Detect Fault Indicate fault to Safe State the System	
	Stop operation		Continue operation Continue degraded Stop operation		Sufficient vehicle level redundancy to continue full operation	
	Rely on driver		Partially rely on driver		No reliance on driver	
	0	1	2	3	4	5

FAIL-SAFE
 DEGRADED MODE
FAIL-OPERATIONAL

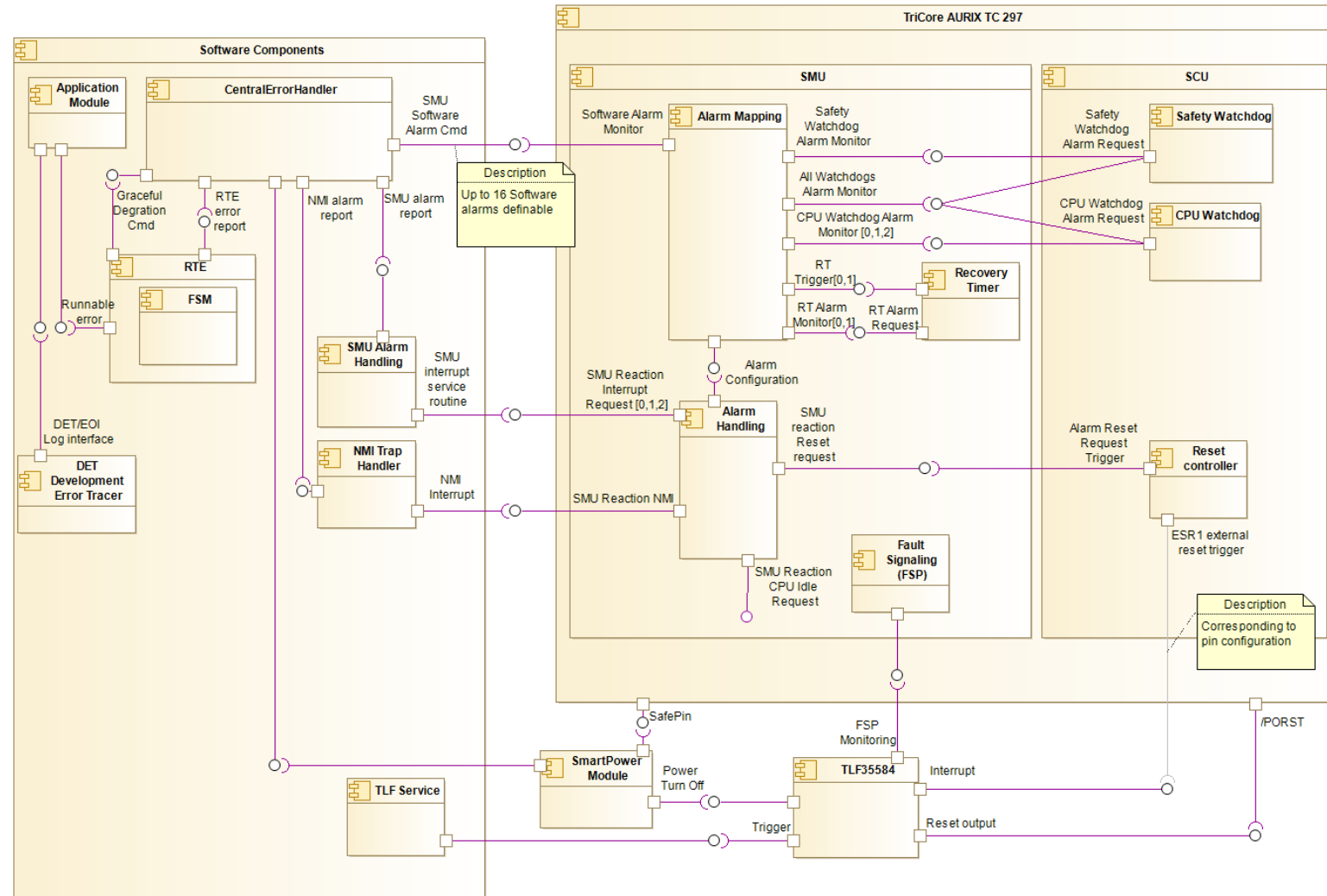
SYSTEM AVAILABILITY

<https://blog.nxp.com/automotive/automotive-functional-safety-the-evolution-of-fail-safe-to-fail-operational-architecture>

Diagnostic Coverage / Monitor versus Redundant Channel



Error Handling on a MCU – Complex Story...



Result of the Hazard and Risk Analysis: A set of Safety Requirements and the required ASIL System Level

Id	Requirements	ASIL _{req}
SR1	The brake signal must be reliable	C
SR2	The gas pedal signal must be reliable	C
SR3	The car may only start after an explicit ignition sequence	QM
SR4	The control speed value must be correctly calculated	C
SR5	The engine currentspeed must be in line with the controlspeed value	C
SR6	The system status must be shown to the driver	A
SR7	The system integrity must be supervised	C

- Functional View
- Hardware View
- Software View



Design Verification

Table 1 — System architectural design analysis

Methods		ASIL			
		A	B	C	D
1	Deductive analysis	o	+	++	++
2	Inductive analysis	++	++	++	++

- inductive analysis methods are bottom-up methods that start from known causes and identify possible effects;
- deductive analysis methods are top-down methods that start from known effects and seek possible causes.

Safety Analysis Methods

Method	Description	Comment
FMEDA (Failure Modes, Effects and Diagnostics Analysis)	Analysis of the effect of random hardware faults on a safety requirement / goal. Including quantitative estimation of failure rates and the resulting probability of a safety goal violation.	- Quantitative - Bottom-Up / Inductive - Hardware
FTA (Fault Tree Analysis)	Top level failures are broken down into a Boolean combination of lower level faults (root causes)	- Qualitative / quantitative - Top Down / Deductive - Hardware / Software
HAZOP (Hazard and Operability Study)	Structures and systematic analysis of a product, using a keyword based approach (e.g. no, more, less, other than,...)	- Qualitative - Top-Down / Deductive - Mostly for System / Software
DFA (Dependent Failure Analysis)	Analysis to identify single events that can cause multiple sub-parts to malfunction.	- Qualitative - Hardware and Software
ETA (Event Tree Analysis)	Systematic analysis of the reaction of a system on events. For every event, the probability for a correct handling as well as for an faulty handling are added.	- Quantitative - Top-Down

Example FMEDA

1	2	J	K	L	M	N	O	P	Q	R	S	T							
	1	Safety Measures				ISO 26262 Metrics													
		Safety Measure to Prevent Violation of Safety Goal	$K_{FMC,RF}$	Safety Measure to Prevent Faults from Being Latent	$K_{FMC,MPF}$	Λ_s	Λ_{RF}	Λ_{SPF}	Λ_{DPFD}	Λ_{DPFL}	SPFM	LFM							
	2																		
	1390	SM1[HW].SRAM:ECC	100.00%	SM1[HW].SRAM:Monitor	90.00%	-	-	-	0.980 FIT	0.109 FIT	100.00%	90.00%							
	1391	SM1[HW].SRAM:ECC	100.00%	SM1[HW].SRAM:Monitor	90.00%	-	-	-	0.782 FIT	0.087 FIT	100.00%	90.00%							
	1392	SM1[HW].SRAM:ECC [FMC=DBE]	100.00%	SM1[HW].SRAM:ECC [FMC=DBE]	100.00%	-	-	-	0.053 FIT	-	100.00%	100.00%							
	1393	SM1[HW].SRAM:ECC [FMC=MBE] & SM1[AoU].SRAM:TEST	60.00%	SM1[HW].SRAM:ECC [FMC=MBE] & SM1[AoU].SRAM:TEST	100.00%	-	0.043 FIT	-	0.064 FIT	-	60.00%	100.00%							
	1447	- SM1[AoU].SRAM:TEST - SM1[AoU].WDT:TIMEOUT - SM1[HW].SRAM:ECC:LOCKSTEP - SM1[HW].SRAM:ADDRMON - SM1[HW].SRAM:ECC - SM1[HW].SRAM:ECC [FMC=ALL-0/1] - SM1[HW].SRAM:ECC [FMC=DBE] - SM1[HW].SRAM:ECC [FMC=MBE]				- SM1[AoU].SRAM:TEST - SM1[AoU].WDT:TIMEOUT - SM1[HW].SRAM:ECC:LOCKSTEP - SM1[HW].SRAM:ADDRMON - SM1[HW].SRAM:ECC [FMC=ALL-0/1] - SM1[HW].SRAM:ECC [FMC=DBE] - SM1[HW].SRAM:ECC [FMC=MBE] - SM1[HW].SRAM:Monitor - SM2[STL].SRAM:ADDRMON - SM2[STL].SRAM:ECC							6.766 FIT	0.125 FIT	0.170 FIT	17.743 FIT	0.933 FIT	98.85%	96.33%

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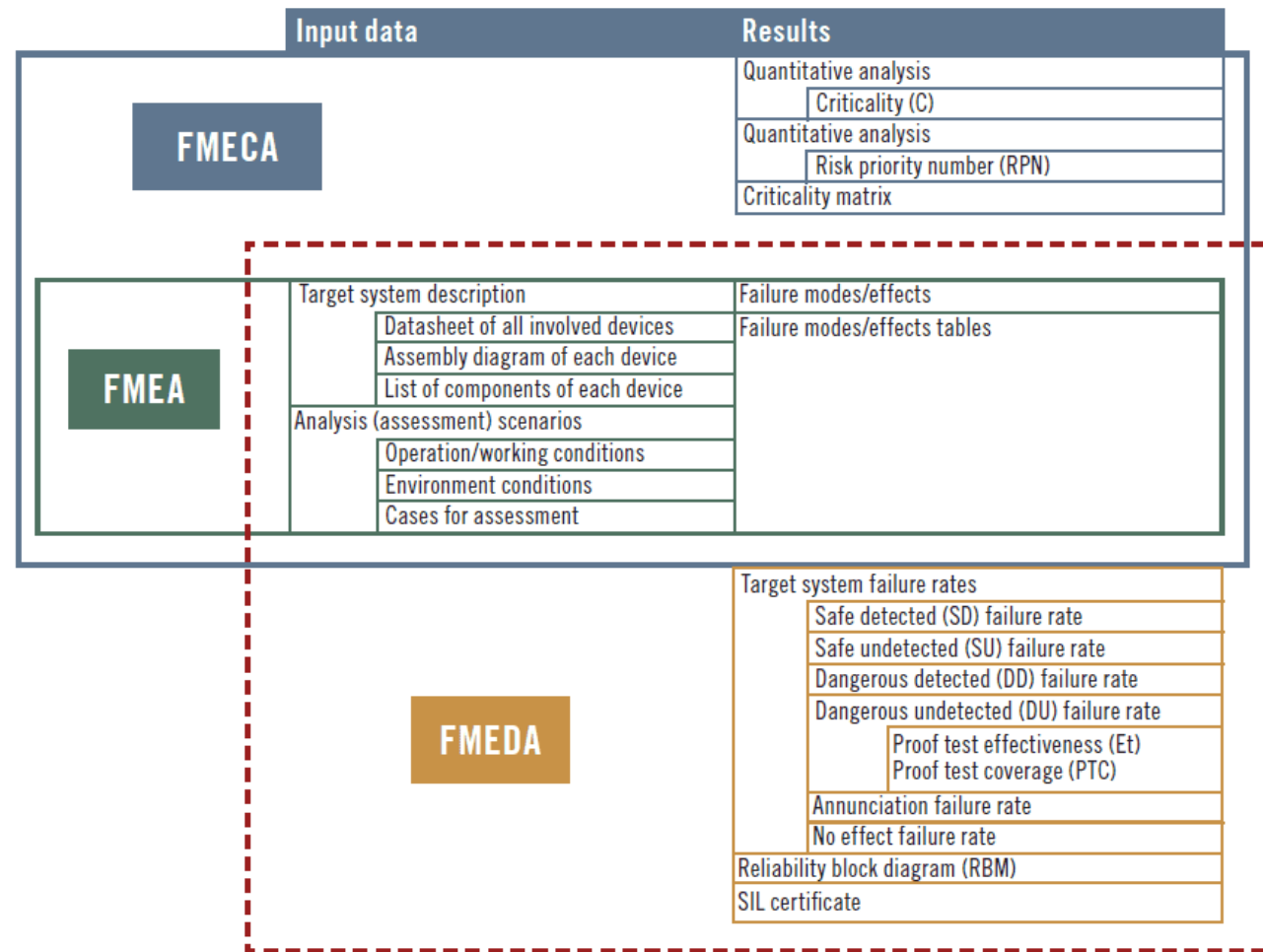
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Operating Profile				
Product Life Time				
Total Life Time	20 yrs			
Total Operating Hours	10,000 hrs			
Time Base for Failure Rates	Operating Time			
Usage Ratio	100.0%			
Avg. Junction Temperature				
Temperature during Operation	125°C			
Temperature during Off-time	40°C			
Note: These temperature values are used for the calculation of failure rates. They should be adjusted to the application's operational profil, but must not exceed the allowed temperature range defined in the data sheet of the microcontroller device.				
FMEDA Results				
ISO 26262 Results		Hard Error (HE)	Soft Error (SE)	Total (HE+SE)
Total Failure Rate	λ	456.503 FIT	4169.408 FIT	4625.911 FIT
Safe Faults ¹⁾	λ_S	9.339 FIT	2069.738 FIT	2079.078 FIT
Single Point and Residual Faults	$\lambda_{SPF} + \lambda_{RF}$	6.085 FIT	6.866 FIT	12.952 FIT
Detected Dual Point Faults ²⁾	$\lambda_{DPF,D}$	401.652 FIT	1888.729 FIT	2290.381 FIT
Latent Dual Point Faults ²⁾	$\lambda_{DPF,L}$	39.427 FIT	204.074 FIT	243.501 FIT
No Part Faults ⁴⁾		17.233 FIT	624.653 FIT	641.886 FIT
Single Point Fault Metric	$SPFM = 1 - (\lambda_{SPF} + \lambda_{RF}) / \lambda$	98.67%	99.84%	99.72%
Latent Fault Metric	$LFM = 1 - \lambda_{DPF,L} / (\lambda - \lambda_{SPF} - \lambda_{RF})$	91.25%	95.10%	94.72%
IEC 61508 Results		Hard Error (HE)	Soft Error (SE)	Total (HE+SE)

FMEA, FMECA, FMEDA

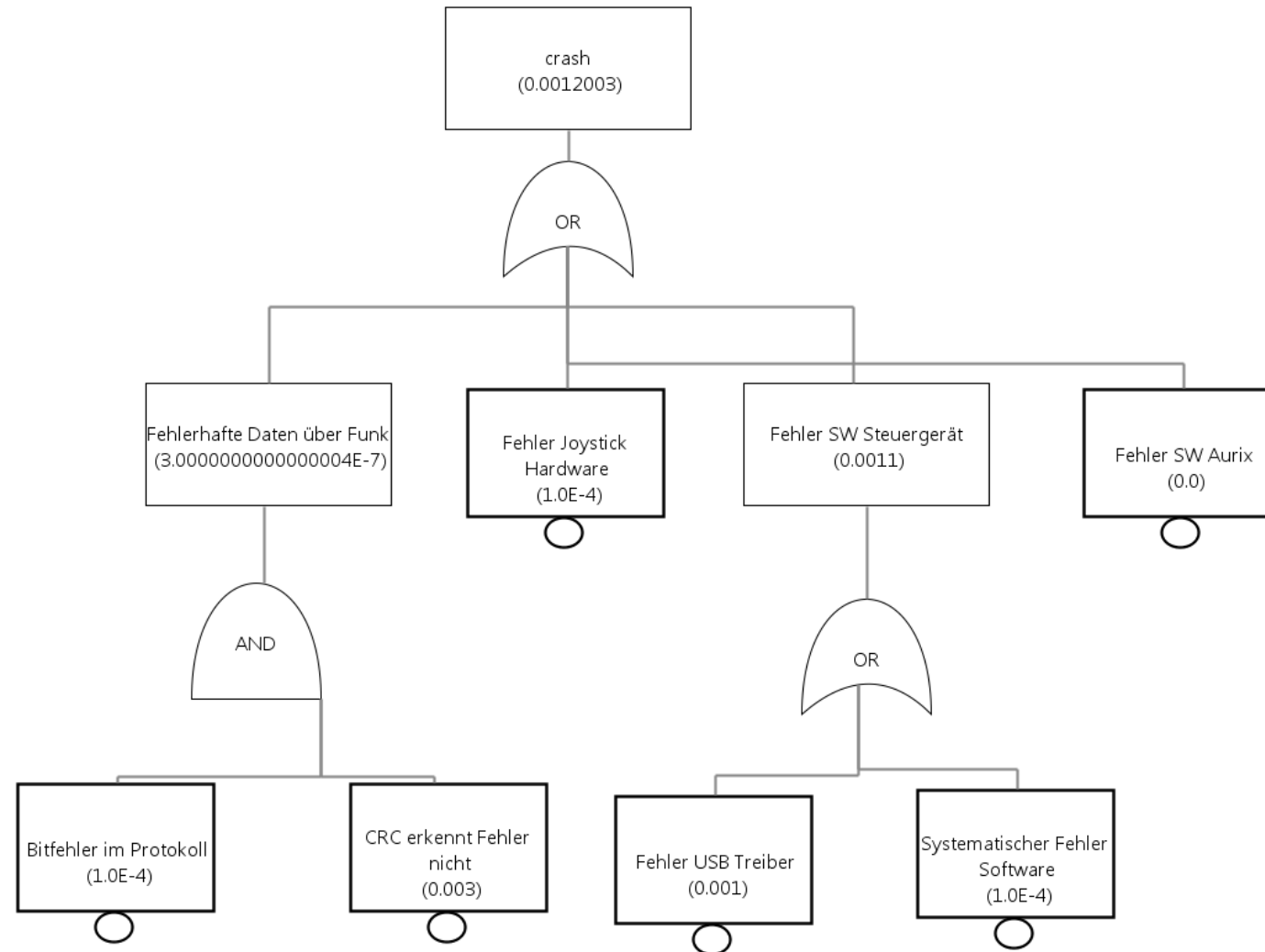


Adds Criticality

Adds Diagnostics

Source: <https://www.controlglobal.com/articles/2018/evaluating-fmea-fmea-and-fmeda/>

Example FTA



Example HAZOP – keyword based analysis to identify potential risks

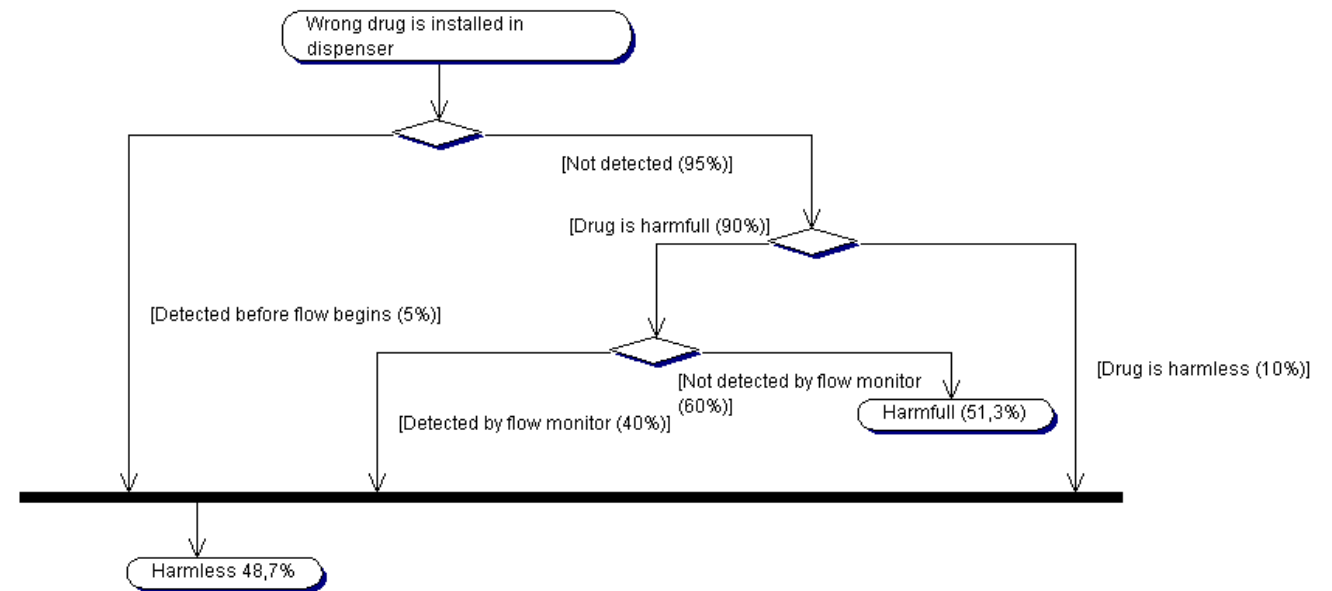
Analysis Focus „Communication Protocol“

- Communication protocol has **WRONG** content → Engine will rotate with wrong speed
 - Possible detection: CRC
- Communication protocol is **NOT** transmitted → Engine speed will not change
 - Possible detection: Sequence Counter

Guide Word	Meaning
NO OR NOT	Complete negation of the design intent
MORE	Quantitative increase
LESS	Quantitative decrease
AS WELL AS	Qualitative modification/increase
PART OF	Qualitative modification/decrease
REVERSE	Logical opposite of the design intent
OTHER THAN / INSTEAD	Complete substitution
EARLY	Relative to the clock time
LATE	Relative to the clock time
BEFORE	Relating to order or sequence
AFTER	Relating to order or sequence

Source: https://en.wikipedia.org/wiki/Hazard_and_operability_study

ETA example



Safety Bias

„People will focus and interpret evidence in a way that confirms the goal they have set for themselves. If the goal is to prove the system is safe, they will focus on the evidence that shows it is safe and create an argument for safety. If the goal is to show that the system is unsafe, the evidence used and the interpretation of available evidence will be quite different. People also tend to interpret ambiguous evidence as supporting their existing position.”

C. Hobbs, *Embedded software development for safety-critical systems*. Boca Raton, FL, London, New York: CRC Press, 2016.

Wrap-Up

- There is no such thing as THE SAFE ARCHITECTURE
- But there are many helpful patterns
 - Reliable Components
 - Hardware: Failure Rate
 - Software: Avoidance of Systematic Errors / Process
 - Fault Detection
 - Diagnostic Coverage
 - Architecture
 - Redundancy
 - Freedom from Interference and Common Cause Failure
 - Safe State